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(12) **United States Patent**
Van Maris et al.(10) **Patent No.: US 12,398,398 B2**(45) **Date of Patent: *Aug. 26, 2025**(54) **RECOMBINANT YEAST EXPRESSING
RUBISCO AND PHOSPHORIBULOKINASE**(71) Applicant: **DANISCO US INC.**, Palo Alto, CA
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Wisselink**, Delft (NL)(73) Assignee: **Danisco US Inc.**, Palo Alto, CA (US)(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.This patent is subject to a terminal dis-
claimer.(21) Appl. No.: **17/108,027**(22) Filed: **Dec. 1, 2020**(65) **Prior Publication Data**

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May 15, 2018, now abandoned, which is a
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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,241,184 A * 12/1980 Hou C12P 7/04
435/150

10,093,937 B2 10/2018 Van Maris et al.

10,689,670 B2 6/2020 Papapetridis et al.

10,982,195 B2 * 4/2021 De Waal C12P 7/06

11,186,850 B2 * 11/2021 Papapetridis C12P 7/06

11,203,741 B2 * 12/2021 De Waal C12Y 101/01072

11,274,310 B2 * 3/2022 De Bruijn C12P 7/06

11,384,368 B2 * 7/2022 De Waal C12Y 401/01039

11,624,057 B2 * 4/2023 De Waal C12Y 207/01028
435/16211,667,886 B2 * 6/2023 Papapetridis C12N 9/0006
435/255.2

2008/0085341 A1 4/2008 Dai et al.

2011/0269180 A1 11/2011 Brat et al.

2012/0064622 A1 3/2012 Fischer

2018/0251773 A1 9/2018 Van Maris et al.

FOREIGN PATENT DOCUMENTS

EP 2958986 12/2015

EP 3318624 A2 5/2018

WO 2008028019 A1 3/2008

WO 2009013159 A2 1/2009

WO 2009/036095 A1 3/2009

WO 2009062190 A2 5/2009

WO 2010057022 A1 5/2010

WO 2011/006126 A2 1/2011

WO 2011010923 A1 1/2011

WO 2014/129898 A2 8/2014

OTHER PUBLICATIONS

Brenda Information on xylitol dehydrogenase, obtained from <https://www.brenda-enzymes.org/enzyme.php?ecno=1.1.1.B19>, Mar. 9, 2022,
3 pages (Year: 2022).*

McKinlay et al., PNAS 107:116699-11675, 2010 (Year: 2010).*

GenBank Database Accession No. AAA99178, Oct. 2011, 2 pages.

UniProt Database Accession No. Q3SFL5, 3 pages (Year: 2013).

UniProt Database Accession No. POA6F7, 2 pages (Year: 2013).

UniProt Database Accession No. POA6G1, 2 pages (Year: 2013).

Milanez et al., Gene 66:55-63, (Year: 1988).

UniProt Database Accession No. P09559, 2 p. (Year: 2013).

Chang et al., J. Biol. Chem. 279:13778-13785, (Year: 2004).

Nissen et al., Met. Engineer. 2:69-77, (Year: 2000).

Letain et al. "Development of a Genetic System for the
Chemolithoautotrophic Bacterium *Thiobacillus denitrificans*", Appl.

(Continued)

Primary Examiner — David Steadman

(57) **ABSTRACT**

The invention relates to a recombinant yeast cell, in particular a transgenic yeast cell, functionally expressing one or more recombinant, in particular heterologous, nucleic acid sequences encoding ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) and phosphoribulokinase (PRK). The invention further relates to the use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism, in particular a eukaryotic micro-organism.

28 Claims, 1 Drawing Sheet**Specification includes a Sequence Listing.**

(56)

References Cited

OTHER PUBLICATIONS

Environ. Microbiol. 73:3265-3271, 2007.

Goloubinoff et al., Nature 337:44-47, 1989.

Chang, H., "Mechanisms of De Novo Multi-domain Protein Folding in Bacteria and Eukaryotes", Dissertation, Ludwig Maximilian University of Munich, 2007.

Mueller-Cajar et al., BioEssays 29:722-724, 2007.

Brandes et al., J. Biol. Chem. 271 :6490-6496, 1996.

English et al., FEMS Microbiol. Lett. 94:111-120, 1992.

D3 Excerpt from GenomeQuest filed on Oct. 21, 2019 in EP Patent Application No. 14708109.

Third Party Observation filed Oct. 31, 2017 in EP Patent Application No. 14708109.

Wikipedia.org entry (Atomosphere of Earth) [downloaded on Jan. 17, 2020].

Kerner, Michael J. et al., "Proteome-wide Analysis of Chaperonin-Dependent Protein Folding in *Escherichia coli*", Cell, Jul. 29, 2005, pp. 209-220, vol. 122.

Guo et al., Comp. Struct. Biotechnol. J. 15:161-167, 2017 (Year: 2017).

Zheng Ping et al., "*Escherichia coli* GroE System and GmE-assisted Rubisco Assembly," Progress in Biotechnology, (2000), vol. 20, No. 3: 18-22.

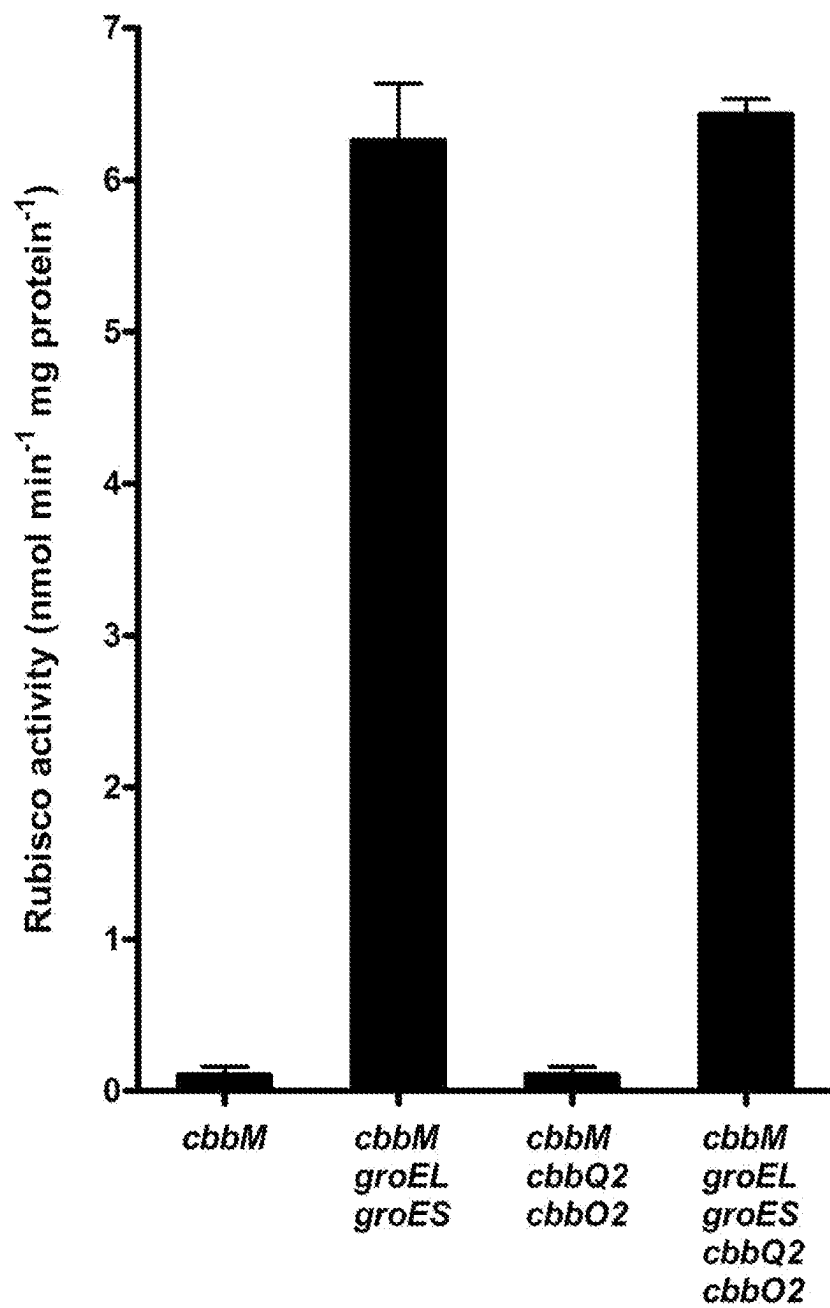
Torben L. Nissen, et al., "Anaerobic and aerobic batch cultivations of *Saccharomyces cerevisiae*," Yeast, (2000), No. 16: 463-474.

Chinese Office Action relating to Chinese Patent Application No. 201480010133.5 dated May 17, 2017.

Goloubinoff, P., et al., Reconstitution of Active Dimeric Ribulose Bisphosphate . . . , Nature, vol. 342, No. 6252, pp. 884-889, 1989.

Tabita, F. R., et al., Function, Structure, and Evolution of the Rubisco-Like . . . , Microbiology and Molecular Biology Reviews 200712 US, vol. 71, No. 4, 2007.

* cited by examiner



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RECOMBINANT YEAST EXPRESSING RUBISCO AND PHOSPHORIBULOKINASE

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is a continuation patent application of U.S. patent application Ser. No. 15/980,413, filed May 15, 2018 is a continuation patent application of U.S. patent application Ser. No. 14/767,661, filed Aug. 13, 2015, now U.S. Pat. No. 10,093,937, issued Oct. 9, 2018, which is a national phase of International Patent Application No. PCT/NL2014/050106, filed Feb. 21, 2014, which claims the benefit of European Patent Application No. 13156448.6, filed Feb. 22, 2013, the disclosures of each of which is incorporated by reference in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED AS A COMPLIANT ASCII TEXT FILE (.txt)

Pursuant to the EFS-Web legal framework and 37 CFR §§ 1.821-825 (see MPEP § 2442.03(a)), a Sequence Listing in the form of an ASCII-compliant text file (entitled “2919208-499002_Sequence_Listing_ST25.txt” created on 18 Nov. 2020, and 66,586 bytes in size) is submitted concurrently with the instant application, and the entire contents of the Sequence Listing are incorporated herein by reference.

FIELD OF THE INVENTION

The invention relates to a recombinant micro-organism having the ability to produce a desired fermentation product, to the functional expression of heterologous peptides in a micro-organism, and to a method for producing a fermentation product wherein said microorganism is used. In a preferred embodiment the micro-organism is a yeast. The invention is further related to a use of CO₂ in micro-organisms.

BACKGROUND OF THE INVENTION

Microbial fermentation processes are applied for industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks.

Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD⁺ can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to reoxidize NADH formed in biosynthetic reactions.

Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology, but various other compounds, including other alcohols, carboxylic acids, isoprenoids, amino acids etc, are currently produced in industrial biotechnological processes.

Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification.

WO 2008/028019 relates to a method for forming fermentation products utilizing a microorganism having at least one heterologous gene sequence, the method comprising the steps of converting at least one carbohydrate to 3-phospho-

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glycerate and fixing carbon dioxide, wherein at least one of said steps is catalyzed by at least one exogenous enzyme. Further, it relates to a microorganism for forming fermentation products through fermentation of at least one sugar, the microorganism comprising at least one heterologous gene sequence encoding at least one enzyme selected from the group consisting of phosphopentose epimerase, phosphoribulokinase, and ribulose biphosphate carboxylase.

In an example, a yeast is mentioned wherein a heterologous PRK and a heterologous Rubisco gene are incorporated. In an embodiment the yeast is used for ethanol production. The results (FIG. 24) show concentrations for transgenic controls and the modified strains. Little difference is noticeable between modified yeast and its corresponding control. No information is apparent regarding product yield, sugar conversion, yeast growth, evaporation rates of ethanol. Thus, it is apparent that results are not conclusive with respect to an improvement in ethanol yield.

Further, WO 2008/028019 is silent on the problem of glycerol side-product formation.

A major challenge relating to the stoichiometry of yeast-based production of ethanol, but also of other compounds, is that substantial amounts of NADH-dependent side-products (in particular glycerol) are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al. Yeast 16 (2000) 463-474). Under conditions that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.

Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed by decarboxylation of pyruvate to ethanol via NAD⁺-dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD⁺ to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD⁺-dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.

In WO 2011/010923, the NADH-related side-product (glycerol) formation in a process for the production of ethanol from a carbohydrate containing feedstock—in particular a carbohydrate feedstock derived from lignocellulosic biomass-glycerol side-production problem is addressed by providing a recombinant yeast cell comprising one or more recombinant nucleic acid sequences encoding an NAD⁺-dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity, said cell either lacking enzymatic activity needed for the NADH-dependent glycerol synthesis

or the cell having a reduced enzymatic activity with respect to the NADH-dependent glycerol synthesis compared to its corresponding wild-type yeast cell. A cell is described that is effective in essentially eliminating glycerol production. Also, the cell uses acetate to reoxidise NADH, whereby ethanol yield can be increased if an acetate-containing feedstock is used.

Although the described process in WO 2011/010923 is advantageous, there is a continuing need for alternatives, in particular alternatives that also allow the production of a useful organic compound, such as ethanol, without needing acetate or other organic electron acceptor molecules in order to eliminate or at least reduce NADH-dependent side-product synthesis. It would in particular be desirable to provide a microorganism wherein NADH-dependent side-product synthesis is reduced and which allows increased product yield, also in the absence of acetate.

The inventors realised that it may be possible to reduce or even eliminate NADH-dependent side-product synthesis by functionally expressing a recombinant enzyme in a heterotrophic, chemotrophic microorganism cell, in particular a yeast cell, using carbon dioxide as a substrate.

SUMMARY OF THE INVENTION

Accordingly, the present invention relates to the use of carbon dioxide as an electron acceptor in a recombinant chemoheterotrophic micro-organism, in particular a eukaryotic micro-organism. Chemotrophic, (chemo)heterotrophic and autotrophic and other classifications of a microorganism are herein related to the micro-organism before recombination, this organism is herein also referred to as the host. For instance, through recombination as disclosed herein a host micro-organism that is originally (chemo)heterotroph and not autotrophic may become autotrophic after recombination, since applying what is disclosed herein causes that the recombined organism may assimilate carbon dioxide, thus resulting in (partial) (chemo)autotrophy.

Advantageously, the inventors have found a way to incorporate the carbon dioxide as a co-substrate in metabolic engineering of heterotrophic industrial microorganisms that can be used to improve product yields and/or to reduce side-product formation.

In particular, the inventors found it to be possible to reduce or even eliminate NADH-dependent side-product synthesis by functionally expressing at least two recombinant enzyme from two specific groups in a eukaryotic microorganism, in particular a yeast cell, wherein one of the enzymes catalysis a reaction wherein carbon dioxide is used and the other uses ATP as a cofactor.

Accordingly, the invention further relates to a recombinant, in a particular transgenic, eukaryotic microorganism, in particular a yeast cell, said microorganism functionally expressing one or more recombinant, in particular heterologous, nucleic acid sequences encoding a ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco) and a phosphoribulokinase (PRK).

BRIEF DESCRIPTION OF THE DRAWING

FIG. 1 is a graph showing comparative results for Rubisco activity in cell extracts described in Example 5.

DETAILED DESCRIPTION OF THE INVENTION

A microorganism according to the invention has in particular been found advantageous in that in the presence of

Rubisco and the PRK NADH-dependent side-product formation (glycerol) is reduced considerably or essentially completely eliminated and production of the desired product can be increased. It is thought that the carbon dioxide acts as an electron acceptor for NADH whereby less NADH is available for the reaction towards the side-product (such as glycerol).

The invention further relates to a method for preparing an organic compound, in particular an alcohol, organic acid or amino acid, comprising converting a carbon source, in particular a carbohydrate or another organic carbon source using a microorganism, thereby forming the organic compound, wherein the microorganism is a microorganism according to the invention or wherein carbon dioxide is used as an electron acceptor in a recombinant chemotrophic or chemoheterotrophic micro-organism.

The invention further relates to a vector for the functional expression of a heterologous polypeptide in a yeast cell, wherein said vector comprises a heterologous nucleic acid sequence encoding Rubisco and PRK, wherein said Rubisco exhibits activity of carbon fixation. The term “a” or “an” as used herein is defined as “at least one” unless specified otherwise.

When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “compound”, this means “at least one” of that moiety, e.g. “at least one compound”, unless specified otherwise.

The term ‘or’ as used herein is to be understood as ‘and/or’.

When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular method of the invention; in particular when referring to such as compound, it includes the natural isomer(s).

For the purpose of clarity and a concise description features are described herein as part of the same or separate embodiments, however, it will be appreciated that the scope of the invention may include embodiments having combinations of all or some of the features described”. In view of this passage it is evident to the skilled reader that the variants of claim 1 as filed may be combined with other features described in the application as filed, in particular with features disclosed in the dependent claims, such claims usually relating to the most preferred embodiments of an invention.

The term ‘fermentation’, ‘fermentative’ and the like is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell, in particular a yeast cell, and usually corresponds to an oxygen consumption of less than 5 mmol/l·h, in particular to an oxygen consumption of less than 2.5 mmol/l·h, or less than 1 mmol/l·h. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This usually corresponds to a dissolved oxygen concentration in the culture broth of less than 5% of air saturation, in particular to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.

The term “yeast” or “yeast cell” refers to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The

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budding yeasts (“true yeasts”) are classified in the order Saccharomycetales, with *Saccharomyces cerevisiae* as the most well known species.

The term “recombinant (cell)” or “recombinant micro-organism” as used herein, refers to a strain (cell) containing nucleic acid which is the result of one or more genetic modifications using recombinant DNA technique(s) and/or another mutagenic technique(s). In particular a recombinant cell may comprise nucleic acid not present in a corresponding wild-type cell, which nucleic acid has been introduced into that strain (cell) using recombinant DNA techniques (a transgenic cell), or which nucleic acid not present in said wild-type is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant (cell)” in particular relates to a strain (cell) from which DNA sequences have been removed using recombinant DNA techniques.

The term “transgenic (yeast) cell” as used herein, refers to a strain (cell) containing nucleic acid not naturally occurring in that strain (cell) and which has been introduced into that strain (cell) using recombinant DNA techniques, i.e. a recombinant cell).

The term “mutated” as used herein regarding proteins or polypeptides means that at least one amino acid in the wild-type or naturally occurring protein or polypeptide sequence has been replaced with a different amino acid, inserted or deleted from the sequence via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., *Molecular Cloning-A Laboratory Manual*, 2nd ed., Vol. 1-3 (1989). The term “mutated” as used herein regarding genes means that at least one nucleotide in the nucleic acid sequence of that gene or a regulatory sequence thereof, has been replaced with a different nucleotide, or has been deleted from the sequence via mutagenesis, resulting in the transcription of a protein sequence with a qualitatively or quantitatively altered function or the knock-out of that gene.

The term “gene”, as used herein, refers to a nucleic acid sequence containing a template for a nucleic acid polymerase, in eukaryotes, RNA polymerase II. Genes are transcribed into mRNAs that are then translated into protein.

The term “nucleic acid” as used herein, includes reference to a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e. g., peptide nucleic acids). A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have

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been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.

The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at the NC-IUBMB website chem.qmul.ac.uk/iubmb/enzyme. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.

If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to a accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found at the National Center for Biotechnology Information (NCBI- NIH) website ncbi.nlm.nih.gov unless specified otherwise.

Every nucleic acid sequence herein that encodes a polypeptide also, by reference to the genetic code, describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.

The term “functional homologue” (or in short “homologue”) of a polypeptide having a specific sequence (e.g. SEQ ID NO: X), as used herein, refers to a polypeptide comprising said specific sequence with the proviso that one or more amino acids are substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion. This functionality may be tested by use of an assay system comprising a recombinant yeast cell comprising an expression vector for the expression of the homologue in yeast, said expression vector comprising a heterologous nucleic acid sequence

operably linked to a promoter functional in the yeast and said heterologous nucleic acid sequence encoding the homologous polypeptide of which enzymatic activity for converting acetyl-Coenzyme A to acetaldehyde in the yeast cell is to be tested, and assessing whether said conversion occurs in said cells. Candidate homologues may be identified by using in silico similarity analyses. A detailed example of such an analysis is described in Example 2 of WO2009/013159. The skilled person will be able to derive there from how suitable candidate homologues may be found and, optionally upon codon (pair) optimization, will be able to test the required functionality of such candidate homologues using a suitable assay system as described above. A suitable homologue represents a polypeptide having an amino acid sequence similar to a specific polypeptide of more than 50%, preferably of 60% or more, in particular of at least 70%, more in particular of at least 80%, at least 90%, at least 95%, at least 97%, at least 98% or at least 99% and having the required enzymatic functionality. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, "identity" also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as determined by the match between strings of such sequences.

Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by "percent identity" or "percent similarity", which is high or low respectively. Although disputed, to indicate "percent identity" or "percent similarity", "level of homology" or "percent homology" are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal, J. B. (1983) An overview of sequence comparison In D. Sankoff and J. B. Kruskal, (ed.), Time warps, string edits and macromolecules: the theory and practice of sequence comparison, pp. 1-44 Addison Wesley). The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman, S. B. and Wunsch, C. D. (1970) J. Mol. Biol. 48, 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOSS package was used (version 2.8.0 or higher, EMBOSS: The European Molecular Biology Open Software Suite (2000) Rice, P. Longden, I. and Bleasby, A. Trends in Genetics 16, (6) pp 276-277, emboss.bioinformatics.nl). For protein sequences, EBLOSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The

optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

Global Homology Definition

The homology or identity is the percentage of identical matches between the two full sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as "IDENTITY".

Longest Identity Definition

The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labeled in the output of the program as "longest-identity".

A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in the sequence listing).

Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called "conservative" amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is aspartagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine. Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. Preferred conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to ser; Arg to lys; Asn to gln or his; Asp to glu; Cys to ser or ala; Gln to asn; Glu to asp; Gly to pro; His to asn or gln; Ile to leu or val; Leu to ile or val; Lys to arg; gln or glu; Met to leu or ile; Phe to met, leu or tyr; Ser to thr; Thr to ser; Trp to tyr; Tyr to trp or phe; and, Val to ile or leu.

Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least

about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity.

Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

“Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

As used herein, “heterologous” in reference to a nucleic acid or protein is a nucleic acid or protein that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a heterologous structural gene is from a species different from that from which the structural gene was derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention.

The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding an enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected yeast cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., *Methods in Yeast Genetics*, Cold Spring Harbor Laboratory (1982) is a well recognized work describing the various methods available to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.

As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene. Typically, a promoter is located in the 5'-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be constitutive, inducible or repressible. If a promoter is an inducible promoter, then the rate of transcription increases in response to an inducing agent.

The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.

The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid sequence that comprises in the 5' to 3' direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region. “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism's genome and is usually circular in nature.

An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism's genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.

By “host cell” is meant a cell which contains a vector and supports the replication and/or expression of the vector. Host cells may be prokaryotic cells such as *E. coli*, or eukaryotic cells such as yeast, insect, amphibian, or mammalian cells. Preferably, host cells are eukaryotic cells of the order of Actinomycetales.

“Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.

The microorganism, preferably is selected from the group of Saccharomycetaceae, such as *Saccharomyces cerevisiae*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum* and *Saccharomyces bayanus*; *Schizosaccharomyces* such as *Schizosaccharomyces pombe*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces octosporus* and *Schizosaccharomyces cryophilus*; *Torula*

laspora such as *Torulaspora delbrueckii*; *Kluyveromyces* such as *Kluyveromyces marxianus*; *Pichia* such as *Pichia stipitis*, *Pichia pastoris* or *pichia angusta*, *Zygosaccharomyces* such as *Zygosaccharomyces bailii*; *Brettanomyces* such as *Brettanomyces intermedius*, *Brettanomyces bruxellensis*, *Brettanomyces anomalus*, *Brettanomyces custersianus*, *Brettanomyces naardenensis*, *Brettanomyces nanus*, *Dekkera bruxellensis* and *Dekkera anomala*; *Metschnikowia*, *Issatchenkia*, such as *Issatchenkia orientalis*, *Kloeckera* such as *Kloeckera apiculata*; *Aureobasidium* such as *Aureobasidium pullulans*.

In a highly preferred embodiment, the microorganism is a yeast cell is selected from the group of Saccharomycetaceae. In particular, good results have been achieved with a *Saccharomyces cerevisiae* cell. It has been found possible to use such a cell according to the invention in a method for preparing an alcohol (ethanol) wherein the NADH-dependent side-product formation (glycerol) was reduced by about 90%, and wherein the yield of the desired product (ethanol) was increase by about 10%, compared to a similar cell without Rubisco and PRK.

The Rubisco may in principle be selected from eukaryotic and prokaryotic Rubisco's.

The Rubisco is preferably from a non-phototrophic organism. In particular, the Rubisco may be from a chemolithoautotrophic microorganism.

Good results have been achieved with a bacterial Rubisco. Preferably, the bacterial Rubisco originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.

The Rubisco may be a single-subunit Rubisco or a Rubisco having more than one subunit. In particular, good results have been achieved with a single-subunit Rubisco.

In particular, good results have been achieved with a form-II Rubisco, more in particular CbbM.

SEQUENCE ID NO: 2 shows the sequence of a particularly preferred Rubisco in accordance with the invention. It is encoded by the cbbM gene from *Thiobacillus denitrificans*. A preferred alternative to this Rubisco, is a functional homologue of this Rubisco, in particular such functional homologue comprising a sequence having at least 80%, 85%, 90% or 95% sequence identity with SEQUENCE ID NO: 2. Suitable natural Rubisco polypeptides are given in Table 1.

TABLE 1

Rubisco polypeptides		
Source	Accession no.	MAX ID (%)
<i>Thiobacillus denitrificans</i>	AAA99178.2	100
<i>Sideroxydans lithotrophicus</i> ES-1	YP_003522651.1	94
<i>Thiothrix nivea</i> DSM 5205	ZP_10101642.1	91
<i>Halothiobacillus neapolitanus</i> c2	YP_003262978.1	90
<i>Acidithiobacillus ferrooxidans</i> ATCC 53993	YP_00220242.1	88
<i>Rhodoferrax ferrireducens</i> T118	YP_522655.1	86
<i>Thiorhodococcus dreswii</i> AZ1	ZP_08824342.1	85
uncultured prokaryote	AGE14067.1	82

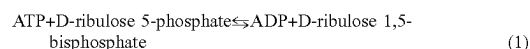
In accordance with the invention, the Rubisco is functionally expressed in the microorganism, at least during use in an industrial process for preparing a compound of interest.

To increase the likelihood that herein enzyme activity is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleotide sequence encoding these enzymes, as well as the Rubisco enzyme and other enzymes of the invention (see

below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleotide sequence encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, 1987, *Nucleic Acids Research* 15: 1281-1295; also see: Jansen et al., 2003, *Nucleic Acids Res.* 31(8):2242-51). An adapted nucleotide sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Most preferred are the sequences which have been codon optimised for expression in the fungal host cell in question such as e.g. *S. cerevisiae* cells.

Preferably, the functionally expressed Rubisco has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent ^{14}C -bicarbonate incorporation by cell extracts of at least $1 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, in particular an activity of at least $2 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, more in particular an activity of at least $4 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$. The upper limit for the activity is not critical. In practice, the activity may be about $200 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less, in particular $25 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, more in particular $15 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less, e.g. about $10 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less. When referred herein to the activity of Rubisco, in particular the activity at 30°C . is meant. The conditions for an assay for determining this Rubisco activity are as found in the Examples, below (Example 4).

A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalysing the chemical reaction:



Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate, whereas its two products are ADP and D-ribulose 1,5-bisphosphate.

PRK belongs to the family of transferases, specifically those transferring phosphorus-containing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. This enzyme participates in carbon fixation.

The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the eukaryotic PRK originates from a plant selected from Caryophyllales, in particular from Amaranthaceae, more in particular from *Spinacia*.

As a preferred alternative to PRK from *Spinacia* a functional homologue of PRK from *Spinacia* may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%, 85%, 90% or 95% sequence identity with SEQUENCE ID NO 4.

Suitable natural PRK polypeptides are given in Table 2.

TABLE 2

Natural PRK polypeptides suitable for expression		
Source	Accession no.	MAX ID (%)
<i>Spinacia oleracea</i>	P09559.1	100
<i>Medicago truncatula</i>	XP_003612664.1	88
<i>Arabidopsis thaliana</i>	NP_174486.1	87
<i>Vitis vinifera</i>	XP_002263724.1	86
<i>Closterium peracerosum</i>	BAL03266.1	82
<i>Zea mays</i>	NP_001148258.1	78

In an advantageous embodiment, the recombinant microorganism further comprises a nucleic acid sequence encoding one or more heterologous prokaryotic or eukaryotic molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the microorganism, in particular with at least one of Rubisco and PRK.

Chaperonins are proteins that provide favourable conditions for the correct folding of other proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine triphosphate (ATP). A review article about chaperones that is useful herein is written by Yebenes (2001); “Chaperonins: two rings for folding”; Hugo Yebenes et al. Trends in Biochemical Sciences, August 2011, Vol. 36, No. 8.

In a preferred embodiment, the chaperone or chaperones are from a bacterium, more preferably from *Escherichia*, in

particular *E. coli* GroEL and GroEs from *E. coli* may in particular encoded in a microorganism according to the invention. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60. If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins.

In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any eukaryotic cell may serve as a chaperone for the Rubisco. See Zeilstra-Ryalls J, Fayet O, Georgopoulos C (1991). “The universally conserved GroE (Hsp60) chaperonins” Annu Rev Microbiol. 45: 301-25. doi:10.1146/annurev.mi.45.100191.001505. PMID 1683763 and Horwich A L, Fenton W A, Chapman E, Farr G W (2007). “Two Families of Chaperonin: Physiology and Mechanism”. Annu Rev Cell Dev Biol. 23: 115-45. doi:10.1146/annurev.cell-bio.23.090506.123555. PMID 17489689.

Particularly good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.

As a preferred alternative to GroEL a functional homologue of GroEL may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%, 85%, 90% or 95% sequence identity with SEQUENCE ID NO: 10.

Suitable natural chaperones polypeptide homologous to SEQUENCE ID NO: 10 are given in Table 3.

TABLE 3

Natural chaperones homologous to SEQUENCE ID NO: 10 polypeptides suitable for expression	
>gi 115388105 ref XP_001211558.1 :2-101 10 kDa heat shock protein, mitochondrial [<i>Aspergillus terreus</i> NIH2624]	
>gi 116196854 ref XP_001224239.1 :1-102 conserved hypothetical protein [<i>Chaetomium globosum</i> CBS 148.51]	
>gi 119175741 ref XP_001240050.1 :3-102 hypothetical protein CIMG_09671 [<i>Coccidioides immitis</i> RS]	
>gi 119471607 ref XP_001258195.1 :12-111 chaperonin, putative [<i>Neosartorya fischeri</i> NRRL181]	
>gi 121699818 ref XP_001268174.1 :8-106 chaperonin, putative [<i>Aspergillus clavatus</i> NRRL 1]	
>gi 126274604 ref XP_001387607.1 :2-102 predicted protein [<i>Scheffersomyces stipitis</i> CBS 6054]	
>gi 146417701 ref XP_001484818.1 :5-106 conserved hypothetical protein [<i>Meyerozyma guilliermondii</i> ATCC 6260]	
>gi 154303611 ref XP_001552212.1 :1-102 10 kDa heat shock protein, mitochondrial [<i>Botryotinia fuckeliana</i> B05.10]	
>gi 156049571 ref XP_001590752.1 :1-102 hypothetical protein SS1G_08492 [<i>Sclerotinia sclerotiorum</i> 1980]	
>gi 156840987 ref XP_001643870.1 :1-103 hypothetical protein Kpol_495p10 [<i>Vanderwaltozyma polyspora</i> DSM 70924]	
>gi 169608295 ref XP_001797567.1 :1-101 hypothetical protein SNOG_07218 [<i>Phaeosphaeria nodorum</i> SN15]	
>gi 171688384 ref XP_001909132.1 :1-102 hypothetical protein [<i>Podospora anserina</i> S mat+]	
>gi 189189366 ref XP_001931022.1 :71-168 10 kDa chaperonin [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]	
>gi 19075598 ref NP_588098.1 :1-102 mitochondrial heat shock protein Hsp10 (predicted) [<i>Schizosaccharomyces pombe</i> 972h-]	
>gi 212530240 ref XP_002145277.1 :3-100 chaperonin, putative [<i>Talaromyces marneffei</i> ATCC 18224]	
>gi 212530242 ref XP_002145278.1 :3-95 chaperonin, putative [<i>Talaromyces marneffei</i> ATCC 18224]	
>gi 213404320 ref XP_002172932.1 :1-102 mitochondrial heat shock protein Hsp10 [<i>Schizosaccharomyces japonicus</i> yFS275]	
>gi 225557301 gb EEH05587.1 :381-478 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> G186AR]	

TABLE 3-continued

Natural chaperones homologous to SEQUENCE ID NO: 10 polypeptides suitable for expression
>gi 225684092 gb EEH22376.1 :3-100 heat shock protein [<i>Paracoccidioides brasiliensis</i> Pb03]
>gi 238490530 ref XP_002376502.1 :2-104 chaperonin, putative [<i>Aspergillus flavus</i> NRRL3357]
>gi 238878220 gb EEQ41858.1 :1-106 10 kDa heat shock protein, mitochondrial [<i>Candida albicans</i> WO-1]
>gi 240280207 gb EER43711.1 :426-523 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> H143]
>gi 241950445 ref XP_002417945.1 :1-103 10 kDa chaperonin, putative; 10 kDa heat shock protein mitochondrial (hsp10), putative [<i>Candida dubliniensis</i> CD36]
>gi 242819222 ref XP_002487273.1 :90-182 chaperonin, putative [<i>Talaromyces stipitatus</i> ATC]
>gi 254566327 ref XP_002490274.1 :1-102 Putative protein of unknown function [<i>Komagataella pastoris</i> GS115]
>gi 254577241 ref XP_002494607.1 :1-103 ZYRO0A05434p [<i>Zygosaccharomyces rouxii</i>]
>gi 255717999 ref XP_002555280.1 :1-103 KLTH0G05588p [<i>Lachancea thermotolerans</i>]
>gi 255956581 ref XP_002569043.1 :2-101 Pc21g20560 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]
>gi 258572664 ref XP_002545094.1 :16-108 chaperonin GroS [<i>Uncinocarpus reesii</i> 1704]
>gi 261190594 ref XP_002621706.1 :3-100 chaperonin [<i>Ajellomyces dermatitidis</i> SLH14081]
>gi 295664909 ref XP_002793006.1 :3-100 10 kDa heat shock protein, mitochondrial [<i>Paracoccidioides</i> sp. 'lutzi' Pb01]
>gi 296412657 ref XP_002836039.1 :76-177 hypothetical protein [<i>Tuber melanosporum</i> Mel28]
>gi 302307854 ref NP_984626.2 :2-102 AEL235Wp [<i>Ashbya gossypii</i> ATCC 10895]
>gi 302894117 ref XP_003045939.1 :1-102 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4]
>gi 303318351 ref XP_003069175.1 :3-100 10 kDa heat shock protein, mitochondrial, putative [<i>Coccidioides posadasii</i> C735 delta SOWgp]
>gi 310795300 gb EFQ30761.1 :1-102 chaperonin 10 kDa subunit [<i>Glomerella graminicola</i> M1.001]
>gi 315053085 ref XP_003175916.1 :12-109 chaperonin GroS [<i>Arthroderma gypseum</i> CBS 118893]
>gi 317032114 ref XP_001394060.2 :334-433 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 317032116 ref XP_001394059.2 :2-101 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 320583288 gb EFW97503.1 :6-106 chaperonin, putative heat shock protein, putative [<i>Ogataea parapolymorpha</i> DL-1]
>gi 320591507 gb EFX03946.1 :1-102 heat shock protein [<i>Grosmannia clavigera</i> kw1407]
>gi 322700925 gb EFY92677.1 :1-102 chaperonin [<i>Metarhizium acridum</i> CQMa 102]
>gi 325096696 gb EGC50006.1 :409-506 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> H88]
>gi 326471604 gb EGD95613.1 :14-111 chaperonin 10 Kd subunit [<i>Trichophyton tonsurans</i> CBS112818]
>gi 327293056 ref XP_003231225.1 :3-100 chaperonin [<i>Trichophyton rubrum</i> CBS 118892]
>gi 330942654 ref XP_003306155.1 :37-136 hypothetical protein PTT_19211 [<i>Pyrenophora teres f. teres</i> 0-1]
>gi 336268042 ref XP_003348786.1 :47-147 hypothetical protein SMAC_01809 [<i>Sordaria macrospora khell</i>]
>gi 340519582 gb EGR49820.1 :1-109 predicted protein [<i>Trichoderma reesei</i> QM6a]
>gi 340960105 gb EGS21286.1 :3-103 putative mitochondrial 10 kDa heat shock protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495]
>gi 342883802 gb EGU84224.1 :1-102 hypothetical protein FOXB_05181 [<i>Fusarium oxysporum</i> Fo5176]
>gi 344302342 gb EGW32647.1 :2-102 hypothetical protein SPAPADRAFT_61712 [<i>Spathaspora passalidarum</i> NRRL Y-27907]
>gi 345570750 gb EGX53571.1 :1-102 hypothetical protein AOL_s00006g437 [<i>Arthrobotrys oligospora</i> ATCC 24927]
>gi 346321154 gb EGX90754.1 :1-102 chaperonin [<i>Cordyceps militaris</i> CM01]
>gi 346970393 gb EGY13845.1 :1-102 heat shock protein [<i>Verticillium dahliae</i> VdLs.17]
>gi 354548296 emb CCE45032.1 :1-106 hypothetical protein CPAR2_700360 [<i>Candida parapsilosis</i>]
>gi 358385052 gb EHK22649.1 :1-102 hypothetical protein TRIVIDRAFT_230640 [<i>Trichoderma virens</i> Gv 29-8]

TABLE 3-continued

Natural chaperones homologous to SEQUENCE ID NO: 10 polypeptides suitable for expression
>gi 358393422 gb EHK42823.1 :1-101 hypothetical protein TRIATDRAFT_258186 [<i>Trichoderma atroviride</i> IMI 206040]
>gi 361126733 gb EHK98722.1 :1-97 putative 10 kDa heat shock protein, mitochondrial [<i>Glare lozoyensis</i> 74030]
>gi 363753862 ref XP_003647147.1 :2-102 hypothetical protein Eeym_5593 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 365758401 gb EHN00244.1 :1-106 Hsp10p [<i>Saccharomyces cerevisiae</i> x <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365987664 ref XP_003670663.1 :1-103 hypothetical protein NDAL_0F01010 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 366995125 ref XP_003677326.1 :1-103 hypothetical protein NCAS_0G00860 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 366999797 ref XP_003684634.1 :1-103 hypothetical protein TPHA_0C00430 [<i>Tetrapisispora phaffii</i> CBS 4417]
>gi 367009030 ref XP_003679016.1 :1-103 hypothetical protein TDEL_0A04730 [<i>Torulaspora delbrueckii</i>]
>gi 367023138 ref XP_003660854.1 :1-104 hypothetical protein MYCTH_59302 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367046344 ref XP_003653552.1 :1-102 hypothetical protein THITE_2116070 [<i>Thielavia terrestris</i> NRRL8126]
>gi 378726440 gb EHY52899.1 :9-109 chaperonin GroES [<i>Exophiala</i> <i>dermatitidis</i> NIH/UT8656]
>gi 380493977 emb CCF33483.1 :1-102 chaperonin 10 kDa subunit [<i>Colletotrichum higginsianum</i>]
>gi 385305728 gb EIF49680.1 :1-102 10 kDa heat shock mitochondrial [<i>Dekkera bruxellensis</i> AWR1499]
>gi 389628546 ref XP_003711926.1 :1-102 hsp10-like protein [<i>Magnaporthe</i> <i>oryzae</i> 70-15]
>gi 396462608 ref XP_003835915.1 :1-101 similar to 10 kDa heat shock protein [<i>Leptosphaeria maculans</i> JN3]
>gi 398392541 ref XP_003849730.1 :1-102 hypothetical protein MYCGRDRAFT_105721 [<i>Zymoseptoria tritici</i> IPO323]
>gi 400597723 gb EJP65453.1 :24-124 chaperonin 10 kDa subunit [<i>Beauveria</i> <i>bassiana</i> ARSEF 2860]
>gi 401623646 gb EJS41738.1 :1-106 hsp10p [<i>Saccharomyces arboricola</i> H-6]
>gi 401842164 gb EJT44422.1 :1-92 HSP10-like protein [<i>Saccharomyces</i> <i>kudriavzevii</i> IFO 1802]
>gi 402084027 gb EJT79045.1 :1-102 hsp10-like protein [<i>Gaeumannomyces</i> <i>graminis</i> var. <i>tritici</i>]
>gi 403215209 emb CCK69709.1 :1-104 hypothetical protein KNAG_0C06130 [<i>Kazachstania naganishii</i> CBS 8797]
>gi 406604629 emb CCH43969.1 :4-100 hypothetical protein BN7_3524 [<i>Wickerhamomyces ciferrii</i>]
>gi 406867021 gb EKD20060.1 :56-156 hypothetical protein MBM_02012 [<i>Marssonina brunnea</i> f. sp. 'multigermutubi' MB_m1]
>gi 407926227 gb EKG19196.1 :74-174 GroES-like protein [<i>Macrophomina</i> <i>phaseolina</i> MS6]
>gi 408398157 gb EKJ77291.1 :11-111 hypothetical protein FPSE_02566 [<i>Fusarium pseudograminearum</i> CS3096]
>gi 410082063 ref XP_003958610.1 :1-103 hypothetical protein KAFR_0H00660 [<i>Kazachstania africana</i> CBS2517]
>gi 425777664 gb EKV15823.1 :58-157 Chaperonin, putative [<i>Penicillium</i> <i>digitatum</i> Pd1]
>gi 440639680 gb ELR09599.1 :1-102 chaperonin GroES [<i>Geomyces</i> <i>destructans</i> 20631-21]
>gi 444323906 ref XP_004182593.1 :1-105 hypothetical protein TBLA_0J00760 [<i>Tetrapisisporablattae</i> CBS 6284]
>gi 448083208 ref XP_004195335.1 :2-101 Piso0_005888 [<i>Milleromyces</i> <i>farinosa</i> CBS 7064]
>gi 448087837 ref XP_004196425.1 :2-102 Piso0_005888 [<i>Milleromyces</i> <i>farinosa</i> CBS 7064]
>gi 448534948 ref XP_003870866.1 :1-106 Hsp10 protein [<i>Candida</i> <i>orthopsilosis</i> Co 90-125]
>gi 449295977 gb EMC91998.1 :1-102 hypothetical protein BAUCODRAFT_39148 [<i>Baudoinia compn</i>]
>gi 46123659 ref XP_386383.1 :3-103 hypothetical protein FG06207.1 [<i>Gibberella zeae</i> PH-1]
>gi 50289455 ref XP_447159.1 :1-103 hypothetical protein [<i>Candida glabrata</i> CBS 138]
>gi 50308731 ref XP_454370.1 :1-103 hypothetical protein [<i>Kluyveromyces</i> <i>lactis</i> NRRL Y-1140]
>gi 50411066 ref XP_457014.1 :1-106 DEHA2B01122p [<i>Debaryomyces</i> <i>hansenii</i> CBS767]
>gi 50545998 ref XP_500536.1 :1-102 YALI0B05610p [<i>Yarrowia lipolytica</i>]
>gi 51013895 gb AAT93241.1 :1-106 YOR020C [<i>Saccharomyces cerevisiae</i>]
>gi 6324594 ref NP_014663.1 :1-106 Hsp10p [<i>Saccharomyces cerevisiae</i> S288c]

TABLE 3-continued

Natural chaperones homologous to SEQUENCE ID NO: 10 polypeptides suitable for expression
>gi 67523953 ref XP_660036.1 :2-101 hypothetical protein AN2432.2 [<i>Aspergillus nidulans</i> FGSC A4]
>gi 70992219 ref XP_750958.1 :12-106 chaperonin [<i>Aspergillus fumigatus</i> Af293]
>gi 85079266 ref XP_956315.1 :1-104 hypothetical protein NCU04334 [<i>Neurospora crassa</i> OR74A]

As a preferred alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%, 85%, 90% or 95% sequence identity with ¹⁵ SEQUENCE ID NO: 12.

Suitable natural chaperones polypeptides homologous to SEQUENCE ID NO: 12 are given in Table 4.

TABLE 4

Natural chaperones homologous to SEQUENCE ID NO: 12 polypeptides suitable for expression
>gi 115443330 ref XP_001218472.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus terreus</i> NIH2624]
>gi 114188341 gb EAU30041.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus terreus</i> NIH2624]
>gi 119480793 ref XP_001260425.1 antigenic mitochondrial protein HSP60, putative [<i>Neosartorya fischeri</i> NRRL 181] >gi 119408579 gb EAW18528.1 antigenic mitochondrial protein HSP60, putative [<i>Neosartorya fischeri</i> NRRL 181]
>gi 126138730 ref XP_001385888.1 hypothetical protein PICST_90190 [<i>Scheffersomyces stipitis</i> CBS 6054] >gi 126093166 gb ABN67859.1 mitochondrial groEL-type heat shock protein [<i>Scheffersomyces stipitis</i> CBS 6054]
>gi 145246630 ref XP_001395564.1 heat shock protein 60 [<i>Aspergillus niger</i> CBS 513.88] >gi 134080285 emb CAK46207.1 unnamed protein product [<i>Aspergillus niger</i>] >gi 350636909 gb EHA25267.1 hypothetical protein ASPNIDRAFT_54001 [<i>Aspergillus niger</i> ATCC 1015]
>gi 146413148 ref XP_001482545.1 heat shock protein 60, mitochondrial precursor [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi 154277022 ref XP_001539356.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1] >gi 150414429 gb EDN09794.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1]
>gi 154303540 ref XP_001552177.1 heat shock protein 60 [<i>Botryotinia fuckeliana</i> B05.10] >gi 347840915 emb CCD55487.1 similar to heat shock protein 60 [<i>Botryotinia fuckeliana</i>]
>gi 156063938 ref XP_001597891.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980] >gi 154697421 gb EDN97159.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980 UF-70]
>gi 156844469 ref XP_001645297.1 hypothetical protein Kpol_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294] >gi 156115957 gb EDO17439.1 hypothetical protein Kpol_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294]
>gi 16416029 emb CAB91379.2 probable heat-shock protein hsp60 [<i>Neurospora crassa</i>] >gi 350289516 gb EGZ70741.1 putative heat-shock protein hsp60 [<i>Neurospora tetrasperma</i> FGSC 2509]
>gi 169626377 ref XP_001806589.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15] >gi 111055053 gb EAT76173.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15]
>gi 169783766 ref XP_001826345.1 heat shock protein 60 [<i>Aspergillus oryzae</i> RIB40] >gi 238493601 ref XP_002378037.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi 83775089 dbj BAE65212.1 unnamed protein product [<i>Aspergillus oryzae</i> RIB40]
>gi 220696531 gb EED52873.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi 391869413 gb EIT78611.1 chaperonin, Cpn60/Hsp60p [<i>Aspergillus oryzae</i> 3.042]
>gi 189190432 ref XP_001931555.1 heat shock protein 60, mitochondrial precursor [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]
>gi 187973161 gb EDU40660.1 heat shock protein 60, mitochondrial precursor [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]
>gi 190348913 gb EDK41467.2 heat shock protein 60, mitochondrial precursor [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi 225554633 gb EEH02929.1 hsp60-like protein [<i>Ajellomyces capsulatus</i> G186AR]
>gi 238880068 gb EEQ43706.1 heat shock protein 60, mitochondrial precursor [<i>Candida albicans</i> WO-1]
>gi 239613490 gb EEQ90477.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> ER-3]
>gi 240276977 gb EER40487.1 hsp60-like protein [<i>Ajellomyces capsulatus</i> H143]

TABLE 4-continued

Natural chaperones homologous to SEQUENCE ID NO: 12 polypeptides suitable for expression
>gi 241958890 ref XP_002422164.1 heat shock protein 60, mitochondrial precursor, putative [<i>Candida dubliniensis</i> CD36] >gi 223645509 emb CAX40168.1 heat shock protein 60, mitochondrial precursor, putative [<i>Candida dubliniensis</i> CD36] >gi 254572906 ref XP_002493562.1 Tetradecameric mitochondrial chaperonin [<i>Komagataella pastoris</i> GS115] >gi 238033361 emb CAY71383.1 Tetradecameric mitochondrial chaperonin [<i>Komagataella pastoris</i> GS115] >gi 254579947 ref XP_002495959.1 ZYRO0C07106p [<i>Zygosaccharomyces rouxii</i>] >gi 238938850 emb CAR27026.1 ZYRO0C07106p [<i>Zygosaccharomyces rouxii</i>] >gi 255712781 ref XP_002552673.1 KLTH0C10428p [<i>Lachancea thermotolerans</i>] >gi 238934052 emb CAR22235.1 KLTH0C10428p [<i>Lachancea thermotolerans</i> CBS 6340] >gi 255721795 ref XP_002545832.1 heat shock protein 60, mitochondrial precursor [<i>Candida tropicalis</i> MYA-3404] >gi 240136321 gb EER35874.1 heat shock protein 60, mitochondrial precursor [<i>Candida tropicalis</i> MYA-3404] >gi 255941288 ref XP_002561413.1 Pc16g11070 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255] >gi 211586036 emb CAP93777.1 Pc16g11070 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255] >gi 259148241 emb CAY81488.1 Hsp60p [<i>Saccharomyces cerevisiae</i> EC1118] >gi 260950325 ref XP_002619459.1 heat shock protein 60, mitochondrial precursor [<i>Clavispora lusitanae</i> ATCC 42720] >gi 238847031 gb EEQ36495.1 heat shock protein 60, mitochondrial precursor [<i>Clavispora lusitanae</i> ATCC 42720] >gi 261194577 ref XP_002623693.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> SLH14081] >gi 239588231 gb EEQ70874.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> SLH14081] >gi 327355067 gb EGE83924.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> ATCC 18188] >gi 296422271 ref XP_002840685.1 hypothetical protein [<i>Tuber melanosporum</i> Mel28] >gi 295636906 emb CAZ84876.1 unnamed protein product [<i>Tuber melanosporum</i>] >gi 296809035 ref XP_002844856.1 heat shock protein 60 [<i>Arthroderma otae</i> CBS 113480] >gi 238844339 gb EEQ34001.1 heat shock protein 60 [<i>Arthroderma otae</i> CBS 113480] >gi 302308696 ref NP_985702.2 AFR155Wp [<i>Ashbya gossypii</i> ATCC 10895] >gi 299790751 gb AAS53526.2 AFR155Wp [<i>Ashbya gossypii</i> ATCC 10895] >gi 374108933 gb AEY97839.1 FAFR155Wp [<i>Ashbya gossypii</i> FDAG1] >gi 302412525 ref XP_003004095.1 heat shock protein [<i>Verticillium albo-atrum</i> VaMs.102] >gi 261356671 gb EEY19099.1 heat shock protein [<i>Verticillium albo-atrum</i> VaMs. 102] >gi 302505585 ref XP_003014499.1 hypothetical protein ARB_07061 [<i>Arthroderma benhamiae</i> CBS 112371] >gi 291178320 gb EFE34110.1 hypothetical protein ARB_07061 [<i>Arthroderma benhamiae</i> CBS 112371] >gi 302656385 ref XP_003019946.1 hypothetical protein TRV_05992 [<i>Trichophyton verrucosum</i> HKI 0517] >gi 291183723 gb EFE39322.1 hypothetical protein TRV_05992 [<i>Trichophyton verrucosum</i> HKI 0517] >gi 302915513 ref XP_003051567.1 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4] >gi 256732506 gb EEU45854.1 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4] >gi 310794550 gb EFQ30011.1 chaperonin GroL [<i>Glomerella graminicola</i> M1.001] >gi 315048491 ref XP_003173620.1 chaperonin GroL [<i>Arthroderma gypseum</i> CBS 118893] >gi 311341587 gb EFR00790.1 chaperonin GroL [<i>Arthroderma gypseum</i> CBS 118893] >gi 320580028 gb EFW94251.1 Tetradecameric mitochondrial chaperonin [<i>Ogataea parapolymorpha</i> DL-1] >gi 320586014 gb EFW98693.1 heat shock protein mitochondrial precursor [<i>Grosmannia clavigera</i> kw1407] >gi 322692465 gb EFY84374.1 Heat shock protein 60 precursor (Antigen HIS-62) [<i>Metarhizium acridum</i> CQMa 102] >gi 322705285 gb EFY96872.1 Heat shock protein 60 (Antigen HIS-62) [<i>Metarhizium anisopliae</i> ARSEF 23] >gi 323303806 gb EGA57589.1 Hsp60p [<i>Saccharomyces cerevisiae</i> FostersB] >gi 323307999 gb EGA61254.1 Hsp60p [<i>Saccharomyces cerevisiae</i> FostersO] >gi 323332364 gb EGA73773.1 Hsp60p [<i>Saccharomyces cerevisiae</i> AWRI796] >gi 326468648 gb EGD92657.1 heat shock protein 60 [<i>Trichophyton tonsurans</i> CBS 112818] >gi 326479866 gb EGE03876.1 chaperonin GroL [<i>Trichophyton equinum</i> CBS 127.97] >gi 330915493 ref XP_003297052.1 hypothetical protein PTT_07333 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1] >gi 311330479 gb EFQ94847.1 hypothetical protein PTT_07333 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1] >gi 336271815 ref XP_003350665.1 hypothetical protein SMAC_02337 [<i>Sordaria macrospora k-hell</i>] >gi 380094827 emb CCC07329.1 unnamed protein product [<i>Sordaria macrospora k-hell</i>] >gi 336468236 gb EGO56399.1 hypothetical protein NEUTE1DRAFT_122948 [<i>Neurospora tetrasperma</i> FGSC 2508] >gi 340522598 gb EGR52831.1 hsp60 mitochondrial precursor-like protein [<i>Trichoderma reesei</i> QM6a] >gi 341038907 gb EGS23899.1 mitochondrial heat shock protein 60-like protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495]

TABLE 4-continued

Natural chaperones homologous to SEQUENCE ID NO: 12 polypeptides suitable for expression
>gi 342886297 gb EGU86166.1 hypothetical protein FOXB_03302 [<i>Fusarium oxysporum</i> Fo5176]
>gi 344230084 gb EGV61969.1 chaperonin GroL [<i>Candida tenuis</i> ATCC 10573]
>gi 344303739 gb EGW33988.1 hypothetical protein SPAPADRAFT_59397 [<i>Spathaspora passalidarum</i> NRRL Y-27907]
>gi 345560428 gb EGX43553.1 hypothetical protein AOL_s00215g289 [<i>Arthrobotrys oligospora</i> ATCC 24927]
>gi 346323592 gb EGX93190.1 heat shock protein 60 (Antigen HIS-62) [<i>Cordyceps militaris</i> CM01]
>gi 346975286 gb EGY18738.1 heat shock protein [<i>Verticillium dahliae</i> VdLs.17]
>gi 354545932 emb CCE42661.1 hypothetical protein CPAR2_203040 [<i>Candida parapsilosis</i>]
>gi 358369894 dbj GAA86507.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus kawachii</i> IFO 4308]
>gi 358386867 gb EHK24462.1 hypothetical protein TRIVIDRAFT_79041 [<i>Trichoderma virens</i> Gv29-8]
>gi 358399658 gb EHK48995.1 hypothetical protein TRIATDRAFT_297734 [<i>Trichoderma atroviride</i> IMI 206040]
>gi 363750488 ref XP_003645461.1 hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 356889095 gb AET38644.1 Hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 365759369 gb EHN01160.1 Hsp60p [<i>Saccharomyces cerevisiae</i> × <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365764091 gb EHN05616.1 Hsp60p [<i>Saccharomyces cerevisiae</i> × <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365985626 ref XP_003669645.1 hypothetical protein NDAL_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 343768414 emb CCD24402.1 hypothetical protein NDAL_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 366995970 ref XP_003677748.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 342303618 emb CCC71399.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 367005154 ref XP_003687309.1 hypothetical protein TPHA_0J00520 [<i>Tetrapispora phaffii</i> CBS 4417] >gi 357525613 emb CCE64875.1 hypothetical protein TPHA_0J00520 [<i>Tetrapispora phaffii</i> CBS 4417]
>gi 367017005 ref XP_003683001.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>] >gi 359750664 emb CCE93790.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>]
>gi 367035486 ref XP_003667025.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 347014298 gb AEO61780.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367055018 ref XP_003657887.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126] >gi 347005153 gb AEO71551.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126]
>gi 378728414 gb EHY54873.1 heat shock protein 60 [<i>Exophiala dermatitidis</i> NIH/UT8656]
>gi 380494593 emb CCF33032.1 heat shock protein 60 [<i>Colletotrichum higginsianum</i>]
>gi 385305893 gb EIF49836.1 heat shock protein 60 [<i>Dekkera bruxellensis</i> AWRI1499]
>gi 389638386 ref XP_003716826.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15] >gi 351642645 gb EHA50507.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15] >gi 440474658 gb ELQ43388.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> Y34] >gi 440480475 gb ELQ61135.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> P131]
>gi 393243142 gb EJD50658.1 chaperonin GroL [<i>Auricularia delicata</i> TFB-10046 SS5]
>gi 396494741 ref XP_003844378.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3] >gi 312220958 emb CBY00899.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3]
>gi 398393428 ref XP_003850173.1 chaperone ATPase HSP60 [<i>Zymoseptoria tritici</i> IPO323] >gi 339470051 gb EGP85149.1 hypothetical protein MYCGRDRAFT_75170 [<i>Zymoseptoria tritici</i> IPO323]
>gi 401624479 gb EJS42535.1 hsp60p [<i>Saccharomyces arboricola</i> H-6]
>gi 401842294 gb EJT44530.1 HSP60-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]
>gi 402076594 gb EJT72017.1 heat shock protein 60 [<i>Gaeumannomyces graminis</i> var. <i>tritici</i> R3-111a-1]
>gi 403213867 emb CCK68369.1 hypothetical protein KNAG_0A07160 [<i>Kazachstania naganishii</i> CBS 8797]
>gi 406606041 emb CCH42514.1 Heat shock protein 60, mitochondrial [<i>Wickerhamomyces ciferrii</i>]
>gi 406863285 gb EKD16333.1 heat shock protein 60 [<i>Marssonina brunnea</i> f. sp. 'multigermutubi' MB_m1]

TABLE 4-continued

Natural chaperones homologous to SEQUENCE ID NO: 12 polypeptides suitable for expression
>gi 407922985 gb EKG16075.1 Chaperonin Cpn60 [<i>Macrophomina phaseolina</i> MS6]
>gi 408399723 gb EKJ78816.1 hypothetical protein FPSE_00959 [<i>Fusarium pseudograminearum</i> CS3096]
>gi 410083028 ref XP_003959092.1 hypothetical protein KAFR_0I01760 [<i>Kazachstania africana</i> CBS 2517] >gi 372465682 emb CCF59957.1 hypothetical protein KAFR_0I01760 [<i>Kazachstania africana</i> CBS 2517]
>gi 444315528 ref XP_004178421.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284] >gi 387511461 emb CCH58902.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284]
>gi 448090588 ref XP_004197110.1 PISO0_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 448095015 ref XP_004198141.1 PISO0_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 359378532 emb CCE84791.1 PISO0_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 359379563 emb CCE83760.1 PISO0_004347 [<i>Millerozyma farinosa</i> CBS 7064]
>gi 448526196 ref XP_003869293.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i> Co 90-125] >gi 380353646 emb CCG23157.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i>]
>gi 46123737 ref XP_386422.1 HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [<i>Gibberella zeae</i> PH-1]
>gi 50292099 ref XP_448482.1 hypothetical protein [<i>Candida glabrata</i> CBS 138] >gi 49527794 emb CAG61443.1 unnamed protein product [<i>Candida glabrata</i>]
>gi 50310975 ref XP_455510.1 hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140] >gi 49644646 emb CAG98218.1 KLLA0F09449p [<i>Kluyveromyces lactis</i>]
>gi 50422027 ref XP_459575.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767] >gi 49655243 emb CAG87802.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767]
>gi 50555023 ref XP_504920.1 YALIOF02805p [<i>Yarrowia lipolytica</i>]
>gi 49650790 emb CAG77725.1 YALIOF02805p [<i>Yarrowia lipolytica</i> CLIB122]
>gi 6323288 ref NP_013360.1 Hsp60p [<i>Saccharomyces cerevisiae</i> S288c]
>gi 123579 sp P19882.1 HSP60_YEAST RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = CPN60; AltName: Full = P66; AltName: Full = Stimulator factor I 66 kDa component; Flags:Precursor
>gi 171720 gb AAA34690.1 heat shock protein 60 (HSP60) [<i>Saccharomyces cerevisiae</i>] >gi 577181 gb AAB67380.1 Hsp60p; Heat shock protein 60 [<i>Saccharomyces cerevisiae</i>] >gi 151941093 gb EDN59473.1 chaperonin [<i>Saccharomyces cerevisiae</i> YJM789] >gi 190405319 gb EDV08586.1 chaperonin [<i>Saccharomyces cerevisiae</i> RM11-1a] >gi 207342889 gb EDZ70518.1 YLR259Cp-like protein [<i>Saccharomyces cerevisiae</i> AWRI1631]
>gi 256271752 gb EEU06789.1 Hsp60p [<i>Saccharomyces cerevisiae</i> JAY291]
>gi 285813676 tpg DAA09572.1 TPA: chaperone ATPase HSP60 [<i>Saccharomyces cerevisiae</i> S288c] >gi 323353818 gb EGA85673.1 Hsp60p [<i>Saccharomyces cerevisiae</i> VL3] >gi 349579966 dbj GAA25127.1 K7_Hsp60p [<i>Saccharomyces cerevisiae</i> Kyokai no. 7] >gi 392297765 gb EIW08864.1 Hsp60p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D] >gi 226279 prf 1504305A mitochondrial assembly factor
>gi 68485963 ref XP_713100.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
>gi 68486010 ref XP_713077.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
>gi 6016258 sp O74261.1 HSP60_CANAL RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = 60 kDa chaperonin; AltName: Full = Protein Cpn60; Flags: Precursor >gi 3552009 gb AAC34885.1 heat shock protein 60 [<i>Candida albicans</i>] >gi 46434552 gb EAK93958.1 heat shock protein 60 [<i>Candida albicans</i> SC5314] >gi 46434577 gb EAK93982.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
>gi 71001164 ref XP_755263.1 antigenic mitochondrial protein HSP60 [<i>Aspergillus fumigatus</i> Af293] >gi 66852901 gb EAL93225.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> Af293]
>gi 159129345 gb EDP54459.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> A1163]
>gi 90970323 gb ABE02805.1 heat shock protein 60 [<i>Rhizophagus intraradices</i>]

In an embodiment, a 10 kDa chaperone from Table 3 is combined with a matching 60 kDa chaperone from table 4 of the same organism genus or species for expression in the host.

For instance: >gi|189189366|ref|XP_001931022.1|:71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis*] expressed together with matching >gi|189190432|ref|XP_001931555.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP].

All other combinations from Table 3 and 4 similarly made with same organism source are also available to the skilled person for expression.

Further, one may combine a chaperone from Table 3 from one organism with a chaperone from Table 4 from another organism, or one may combine GroES with a chaperone from Table 3, or one may combine GroEL with a chaperone from Table 4.

As follows from the above, the invention further relates to a method for preparing an organic compound comprising converting a carbon source, using a microorganism, thereby forming the organic compound. The method may be carried out under aerobic, oxygen-limited or anaerobic conditions.

The invention allows in particular a reduction in formation of an NADH dependent side-product, especially glycolic acid.

erol, by up to 100%, up to 99%, or up to 90%, compared to said production in a corresponding reference strain. The NADH dependent side-product formation is preferably reduced by more than 10% compared to the corresponding reference strain, in particular by at least 20%, more in particular by at least 50%. NADH dependent side-product production is preferably reduced by 10-100%, in particular by 20-95%, more in particular by 50-90%.

In preferred method wherein Rubisco, or another enzyme capable of catalysing the formation of an organic compound from CO₂ (and another substrate) or another enzyme that catalyses the function of CO₂ as an electron acceptor, is used, the carbon dioxide concentration in the reaction medium is at least 5% of the CO₂ saturation concentration under the reaction conditions, in particular at least 10% of said CO₂ saturation concentration, more in particular at least 20% of said CO₂ saturation concentration. This is in particular advantageous with respect to product yield. The reaction medium may be oversaturated in CO₂ concentration, saturated in CO₂ concentration or may have a concentration below saturation concentration. In a specific embodiment, the CO₂ concentration is 75% of the saturation concentration or less, in particular 50% of said saturation concentration or less, more in particular is 25% of the CO₂ saturation concentration or less.

In a specific embodiment, the carbon dioxide or part thereof is formed in situ by the microorganism. If desired, the method further comprises the step of adding external CO₂ to the reaction system, usually by aeration with CO₂ or a gas mixture containing CO₂, for instance a CO₂/nitrogen mixture. Adding external CO₂ in particular is used to (increase or) maintain the CO₂ within a desired concentration range, if no or insufficient CO₂ is formed in situ.

Determination of the CO₂ concentration in a fluid is within the routine skills of the person skilled in the art. In practice, one may routinely determine the CO₂ concentration in the gas phase above a culture of the yeast (practically the off-gas if the medium is purged with a gas). This can routinely be measured using a commercial gas analyser, such as a RosemountNGA200000 gas analyser (Rosemount Analytical, Orrville, USA). The concentration in the liquid phase (relative to the saturation concentration), can then be calculated from the measured value in the gas, from the CO₂ saturation concentration and Henri coefficients of under the existing conditions in the method. These parameters are available from handbooks or can be routinely determined.

As a carbon source, in principle any carbon source that the microorganism can use as a substrate can be used. In particular an organic carbon source may be used, selected from the group of carbohydrates and lipids (including fatty acids). Suitable carbohydrates include monosaccharides, disaccharides, and hydrolysed polysaccharides (e.g. hydrolysed starches, lignocellulosic hydrolysates). Although a carboxylic acid may be present, it is not necessary to include a carboxylic acid such as acetic acid, as a carbon source.

It is in particular an advantage of the present invention that an improved ethanol yield and a reduced glycerol production is feasible compared to, e.g., a wild type yeast cell, without needing to intervene in the genome of the cell by inhibition of a glycerol 3-phosphate phosphohydrolase and/or encoding a glycerol 3-phosphate dehydrogenase gene.

Still, in a specific embodiment, a yeast cell according to the invention may comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase and/or encoding a glycerol 3-phosphate dehydrogenase gene:

Herein in the cell, enzymatic activity needed for the NADH-dependent glycerol synthesis is reduced or deleted. The reduction or deleted of this enzymatic activity can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase activity (GPD) or one or more genes encoding a glycerol phosphate phosphatase activity (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encoded a polypeptide with reduced activity.

Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO 2011/010923, and are disclosed in SEQ ID NO: 24-27 of that application. The contents of this application are incorporated by reference, in particular the contents relating to GPD and/or GPP.

As shown in the Examples below, the invention is in particular found to be advantageous in a process for the production of an alcohol, notably ethanol. However, it is contemplated that the insight that CO₂ can be used as an electron acceptor in microorganisms that do not naturally allow this, has an industrial benefit for other biotechnological processes for the production of organic molecules, in particular organic molecules of a relatively low molecular weight, particularly organic molecules with a molecular weight below 1000 g/mol. The following items are mentioned herein as preferred embodiments of the use of carbon dioxide as an electron acceptor in accordance with the invention.

1. Use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism is a non-phototrophic eukaryotic micro-organism.
2. Use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism, wherein the micro-organism produces an organic compound under anaerobic conditions.
3. Use according to item 1 or 2, wherein the carbon dioxide serves as an electron acceptor in a process with NADH as an electron donor.
5. Use according to any of the preceding items, wherein the micro-organism produces an organic compound in a process with an excess production of ATP and/or NADH.
6. Use according to any of the preceding items, wherein the micro-organism comprises a heterologous nucleic acid sequence encoding a polypeptide from a (naturally) autotrophic organism.
7. Use according to item 6, wherein the micro-organism comprises a heterologous nucleic acid sequence encoding a first prokaryotic chaperone for said polypeptide and preferably a nucleic acid sequence encoding a second prokaryotic chaperone—different from the first—for said polypeptide.
8. Use according to item 7, wherein the chaperones are GroEL and GroES.
9. Use according to any of the preceding items, wherein the micro-organism produces an organic compound selected from the group consisting of alcohols (such as methanol, ethanol, propanol, butanol, phenol, polyphenol), ribosomal peptides, antibiotics (such as penicillin), bio-diesel, alkynes, alkenes, isoprenoids, esters,

carboxylic acids (such as succinic acid, citric acid, adipic acid, lactic acid), amino acids, polyketides, lipids, and carbohydrates.

10. Use according to any of the preceding items, wherein the microorganism comprises a heterologous nucleic acid sequence functionally expressing a polypeptide selected from the group consisting of carbonic anhydrases, carboxylases, oxygenases, hydrogenases, dehydrogenases, isomerases, aldolases, transketolases, transaldolases, phosphatases, epimerases, kinases, carboxykinases, oxidoreductases, aconitases, fumarases, reductases, lactonases, phosphoenolpyruvate (PEP) carboxylases, phosphoglycerate kinases, glyceraldehyde 3-phosphate dehydrogenases, triose phosphate isomerases, fructose-1,6-bisphosphatases, sedoheptulose-1,7-bisphosphatases, phosphopentose isomerases, phosphopentose epimerase, phosphoribulokinases (PRK), glucose 6-phosphate dehydrogenases, 6-phosphogluconolactonases, 6-phosphogluconate dehydrogenases, ribulose 5-phosphate isomerases, ribulose 5-phosphate 3-epimerases, Ribulose-1,5-bisphosphate carboxylase oxygenases, lactate dehydrogenases, malate synthases, isocitrate lyases, pyruvate carboxylases, phosphoenolpyruvate carboxykinases, fructose-1,6-bisphosphatases, phosphoglucoisomerases, glucose-6-phosphatases, hexokinases, glucokinases, phosphofructokinases, pyruvate kinases, succinate dehydrogenases, citrate synthases, isocitrate dehydrogenases, α -ketoglutarate dehydrogenases, succinyl-CoA synthetases, malate dehydrogenases, nucleoside-diphosphate kinases, xylose reductases, xylitol dehydrogenases, xylose isomerases, isoprenoid synthases, and xylonate dehydratases.

11. Use according to item 10, wherein the microorganism comprises a heterologous nucleic acid sequence functionally expressing Ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco) and/or a heterologous nucleic acid sequence functionally expressing a phosphoribulokinase (PRK).

12. Use according to any of the preceding items, wherein the microorganism is selected from the group of is selected from the group consisting of Saccharomycetales, *Penicillium*, *Yarrowia* and *Aspergillus*.

13. Use according to any of the preceding items, wherein the carbon dioxide is used as an electron acceptor to reduce production of an NAD⁺-dependent side-product or NADH-dependent side-product, such as glycerol, in a process for preparing another organic compound, such as another alcohol or a carboxylic acid.

14. Recombinant micro-organism, in particular a eukaryotic micro-organism, having an enzymatic system

allowing the micro-organism to use carbon dioxide as an electron acceptor under chemotrophic (non-phototrophic) conditions, wherein the microorganism is preferably as defined in the provision items.

15. Recombinant micro-organism according to item 14, wherein the micro-organism has an enzymatic system for producing an organic compound in a process with an excess production of ATP and/or NADH.

The production of the organic compound of interest may take place in a organism known for its usefulness in the production of the organic compound of interest, with the proviso that the organism has been genetically modified to enable the use of carbon dioxide as an electron acceptor in the organism.

Although it is contemplated that the invention is interesting for the production of a variety of industrially relevant organic compounds, a method or use according to the invention is in particular considered advantageous for the production of an alcohol, in particular an alcohol selected from the group of ethanol, n-butanol and 2,3-butanediol; or in the production of an organic acid/carboxylate, in particular a carboxylate selected from the group of L-lactate, 3-hydroxypropionate, D-malate, L-malate, succinate, citrate, pyruvate and itaconate.

Regarding the production of ethanol, details are found herein above, when describing the yeast cell comprising PRK and Rubisco and in the examples. The ethanol or another alcohol is preferably produced in a fermentative process.

For the production of several organic acids (carboxylates), e.g. citric acid, an aerobic process is useful. For citric acid production for instance *Aspergillus niger*, *Yarrowia lipolytica*, or another known citrate producing organism may be used.

An example of an organic acid that is preferably produced anaerobically is lactic acid. Various lactic acid producing bacterial strains and yeast strains that have been engineered for lactate production are generally known in the art.

EXAMPLES

Example 1. Construction of the Expression Vector

Phosphoribulokinase (PRK) cDNA from *Spinacia oleracea* (spinach) (EMBL accession number: X07654.1) was PCR-amplified using Phusion Hot-start polymerase (Finnzymes, Landsmeer, the Netherlands) and the oligonucleotides XbaI_prk-FW2 and RV1_XhoI_prk (Table 5), and was ligated in pCR®-Blunt II-TOPO® (Life Technologies Europe BV, Bleiswijk, the Netherlands).

TABLE 5

Oligonucleotides				
SEQ ID				
NO	Name	Sequence (5' to 3')	Purpose	
Cloning				
12	XbaI_prk_FW2	TGACATCTAGATGTCACAACAACAAACAATTG	cloning of PRK into pUDE046.	
13	RV1 XhoI prk	TGACATCTAGATGTCACAACAACAAACAATTG	cloning of PRK into pUDE046.	

TABLE 5-continued

Oligonucleotides			
SEQ ID NO	Name	Sequence (5' to 3')	Purpose
Primers used for in vivo plasmid assembly			
14	HR-cbbM-FW-65	TTGTAACACGACGGCCAGTGTAGCGCGCGTAATACG ACTCACTATAGGGCGAATTGGGTACAGCTGGAGCT CAGTTTATCATTATC	Rubisco cbbM cassette for plasmids pUDC075, pUDC099, and pUDC100.
15	HR-cbbM-RV-65	GGAATCTGTGTAGTATGCCTGGAATGTCTGCCGTG CCATAGCCATGTATGTGATATGTCGGTACCGGCC GCAAATTAAAG	Rubisco cbbM cassette for plasmids pUDC075, pUDC099, and pUDC100
16	linker-cbb02-pRS416	ATCACTCTTACCAGGCTAGGACGACCCTACTCATG TATTGAGATCGACGAGATTTCTAGGCCAGCTTTTGT TCCCTTTAGTGAGGTTAATTGCGCGCTTGGCGTAA TCATGGTCATAGC	Linker fragment for assembly of plasmid pUDC099.
17	linker-cbbM-GroEL	GACATATCAGCATACATGGCTATGGCAGCGCAGAC ATTCCAGGCATACATACAGATTCATCACTCTTAC CAGGCTAGGACGACCCTACTCATGTATTGAGATCG ACGAGATTTCTAGG	Linker fragment for assembly of plasmid pUDC100.
Primers used for in vivo integration assembly			
18	FW pTDH3-HR-CAN1up	GTTGGATCCAGTTTTTAATCTGTCGTCAATCGAAAG TTTTTTTCAGAGTTCTTCAGACTTCTTAATCCTGT AAAAACAAAAAAGGCATAGCAAGCTGG AGCTCAGTTTATC	1 st cloning expression cassette linker fragment between CAN1 upstream and PRK expression cassette (IMI229), and CAN1up-linker and KLEU2 expression cassette (IMI232).
19	RV linker-iHR2B	AGATATACTGCAAGTCCGGAGCAACAGTCGTATA ACTCGAGCAGCCCTCTACTTTGTTGTCGCTAAGA GAATGGACC	1 st cloning fragment: linker fragment between CAN1up-linker and PRK expression cassette (IMI229).
20	RV linker-iHR6	GCTATGACCATGATTACGCCAAGCGCAATTAAC CCTCACTAAGGGGAACAAAGCTGGTTGCGCTAAG AGAATGGACC	1 st cloning fragment: linker fragment between CAN1up-linker and KLEU2 expression cassette (IMI232).
21	FW pGAL1-prk HR2B	CAACAAAGTAGAGGGCTGCTCGAGTTATACGACTG TTGCTCCGACTTTTGCAGTATATCTGCTGGAGCTCT AGTACGGATT	2 nd cloning fragment: GAL1 _p -PRK-CYC1, expression cassette (IMI229) from pUDE046.
22	RV CYC1t-prk HR2	GGAATCTGTGTAGTATGCCTGGAATGTCTGCCGTG CCATAGCCATGTATGTGATATGTCGTACCGGCCG CAAATTAAAG	2 nd cloning fragment: GAL1 _p -PRK-CYC1, expression cassette (IMI229) from pUDE046.
23	FW HR2-cbbQ2-HR3	GACATATCAGCATACATGGCTATGG	3 rd cloning fragment: PGII _p -cbbQ2-TEF2, cassette (IMI229).
24	RV HR2-cbbQ2-HR3	GGACACGCTTGACAGAATGTCAAAGG	3 rd cloning fragment: PGII _p -cbbQ2-TEF2, cassette (IMI229).
25	FW HR3-cbb02-HR4	CGTCCGATATGATCTGATTGG	4 th TARI cloning fragment: PGK1 _p -cbb02-ADH1, cassette (IMI229).
26	RV HR3-cbb02-HR4	CCTAGAAATCTCGTCGATCTC	4 th cloning fragment: PGK1 _p -cbb02-ADH1, cassette (IMI229).
27	FW HR4-GroEL-HR5	ATCACTCTTACCAGGCTAGG	5 th cloning fragment: TEF1 _p -groEL-ACT1, cassette (IMI229).
28	RV HR4-GroEL-HR5	CTGGACCTTAATCGTGTGCGCATCCTC	5 th cloning fragment: TEF1 _p -groEL-ACT1, cassette (IMI229).
29	FW HRS-GroES-HR6	CCGTATAGCTTAATAGCCAGCTTTATC	6 th cloning fragment: TPII _p -groES-PGII, cassette (IMI229).
30	RV HRS-GroES-HR6	GCTATGACCATGATTACGCCAAGC	6 th cloning fragment: TPII _p -groES-PGII, cassette (IMI229).
31	FW HR6-LEU2-CAN1dwn	CCAGCTTTTGTTCCTTTAGTGAGGGTTAATTGCGC GCTTGGCGTAATCATGGTCATAGCCTGTGAAGATC CCAGCAAAG	7 th (IMI229) or 2 nd (IMI232) cloning fragment: KLEU2 cassette from pUG73.

TABLE 5-continued

Oligonucleotides			
SEQ ID NO	Name	Sequence (5' to 3')	Purpose
32	RV LEU2 HR-CAN1	AGCTCATGATCCCTTAACTTTCTTTTCGGTGTAT GACTTATGAGGGTGAGAATGCGAAATGGCGTGGA AATGTGATCAAAGGTAATAAACGTCATATATCCG CAGGCTAACCGGAAC	7 th (IMI229) or 2 nd (IMI232) cloning fragment: KLEU2 cassette from pUG73.
Primers used for verification of the in vivo assembled constructs			
33	m-PCR-HR1-FW	GGCGATTAAGTTGGGTAACG	Diagnostic for assembly of plasmids pUDC075, pUDC099, and pUDC100, .
34	m-PCR-HR1-RV	AACTGAGCTCCAGCTGTACC	Diagnostic for assembly of plasmids pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
35	m-PCR-HR2-FW	ACGCGTGTACGCATGTAAC	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229
36	m-PCR-HR2-RV	GCGCGTGGCTTCCTATAATC	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229
37	m-PCR-HR3-FW	GTGAATGCTGGTCGTATAC	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
38	m-PCR-HR3-RV	GTAAGCAGCAACACCTTCAG	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
39	m-PCR-HR4-FW	ACCTGACCTACAGGAAAGAG	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
40	m-PCR-HR4-RV	TGAAGTGGTACGGCGATGC	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
41	m-PCR-HR5-FW	ATAGCCACCCAAGGCATTTT	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
42	m-PCR-HR5-RV	CCGCACTTTCTCCATGAGG	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
43	m-PCR-HR6-FW	CGACGGTTACGGTGTAAAG	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.
44	m-PCR-HR6-RV	CTTCCGGCTCCTATGTTGTG	Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.

After restriction by XbaI and XhoI, the PRK-containing fragment was ligated into pTEF424. The TEF1p was later replaced by GAL1p from plasmid pSH47 by XbaI and SacI restriction/ligation, creating plasmid pUDE046 (see Table 6).

TABLE 6

Plasmids		
Name	Relevant genotype	Source/reference
pFL451	pAOX1-prk (Spinach)-AOX1t (PHIL2-D2 HIS4 Amp centromeric)	Brandes et al. 1996. ¹⁴
pCR ®-Blunt II-TOPO	bla	Life Technologies Europe BV
pTEF424__TEF	TRP1 2μ bla	Mumberg et al. 1995 ²⁵ .
pSH47	URA3 CEN6 ARS4 GAL1 _p -cre-CYC1 _t bla	Güldener et al. 1996 ²⁶
pUD0E46	TRP1 2μ GAL1p-prk-CYC1 _t bla	This study.
pPCR-Script	bla	Life Technologies Europe BV
pGPD__426	URA3 2μ bla	Mumberg et al. 1995 ²⁵ .
pRS416	URA3 CEN6 ARS4 bla	Mumberg et al. 1995 ²⁵ .
pBTWW002	URA3 2μ TDH3 _p -cbbM-CYC1 _t bla	This study.
pUDC098	URA3 CEN6 ARS4 TDH3 _p -cbbM-CYC1 _t bla	This study.
pMK-RQ	nptII	Life Technologies Europe BV
pUD230	PGI1 _p -cbbQ2-TEF2 _t nptII	Life Technologies Europe BV
pUD231	PGK1 _p -cbbO2-ADH1 _t nptII	Life Technologies Europe BV
pUD232	TEF1 _p -groEL-ACT1 _t nptII	Life Technologies Europe BV
pUD233	TPI1 _p -groES-PGI1 _t nptII	Life Technologies Europe BV
pUDC075	URA3 CEN6 ARS4 TDH3 _p -cbbM-CYC1 _t PGI1 _p -cbbQ2-TEF2 _t PGK1 _p -cbbO2-ADH1 _t TEF1 _p -groEL-ACT1 _t TPI1 _p -groES-PGI1 _t bla	This study.
pUDC099	URA3 CEN6 ARS4 TDH3 _p -cbbM-CYC1 _t PGI1 _p -cbbQ2-TEF2 _t PGK1 _p -cbbO2-ADH1 _t bla	This study.
pUDC100	URA3 CEN6 ARS4 TDH3 _p -cbbM-CYC1 _t TEF1 _p -groEL-ACT1 _t TPI1 _p -groES-PGI1 _t bla	This study.

Rubisco form II gene cbbM from *Thiobacillus denitrificans* (*T. denitrificans*) flanked by KpnI and SacI sites was codon optimized synthesized at GeneArt (Life Technologies Europe BV), and ligated into pPCR-Script., the plasmid was then digested by BamHI and SacI. The cbbM-containing fragment was ligated into the BamHI and SacI restricted vector pGPD_426 creating plasmid pBTWW002. The cbbM expression cassette was transferred into pRS416 using KpnI and SacI, yielding pUDC098.

Expression cassette of the specific Rubisco form II chepa-
rones from *T. denitrificans* cbbQ2 and cbbO2, and chaper-
ones groEL and groES from *E. coli*, were codon optimized.
The expression cassettes contained a yeast constitutive pro-
motors and terminator, flanking the codon optimized gene.
The cassette was flanked by unique 60 bp regions obtained
by randomly combining bar-code sequences used in the
Saccharomyces Genome Deletion Project and an EcoRV site
(GeneArt). The expression cassettes were inserted in plas-
mid pMK-RQ (GeneArt) using the SfiI cloning sites yielding

pUB230 (PGI1p-cbbQ2-TEF2t), pUD231 (PGK1p-cbbO2-
ADH1t), pUD232 (TEF1p-groEL-ACT1t), and pUDE233
(TPI1p-groES-PGI1t) Table 6). The expression cassette
TDH3p-cbbM-CYC1t was PCR-amplified from plasmid
pBTWW002 using Phusion Hot-Start Polymerase
(Finnzymes) and primers HR-cbbM-FW-65 and HR-cbbM-
RV-65 in order to incorporate the 60-bp region for recom-
bination cloning.

Example 2. Strain Construction, Isolation and Maintenance

All *Saccharomyces cerevisiae* strains used (Table 7)
belong to the CEN.PK family. All strains were grown in 2%
w/v glucose synthetic media supplemented with 150 mg L⁻¹
uracil when required until they reached end exponential
phase, then sterile glycerol was added up to ca. 30% v/v and
aliquots of 1 ml were stored at -80° C.

TABLE 7

<i>Saccharomyces cerevisiae</i> strains		
Strain	Relevant genotype	Source/reference
CEN.PK113-5D	MATa ura3-52	Euroscarf
CEN.PK102-3A	MATa ura3-52 leu2-3, 112	Euroscarf
IMC014	MATa ura3-52 pUDC075 (CEN6 ARS4 URA3 TDH3 _p - cbmM-CYC1 _i , PGI1 _p -cbmQ2-TEF2 _i , PGK1 _p -cbmO2- ADH1 _i , TEF1 _p -groEL-ACT1 _i , TPI1 _p -groES-PGI1 _i)	This study.
IMC033	MATa ura3-52 pUDC098 (CEN6 ARS4 URA3 TDH3 _p - cbmM-CYC1 _i)	This study.
IMC034	MATa ura3-52 pUDC099 (CEN6 ARS4 URA3 TDH3 _p - cbmM-CYC1 _i , PGI1 _p -cbmQ2-TEF2 _i , PGK1 _p -cbmO2- ADH1 _i , cbmO2-pRS416 linker)	This study.
IMC035	MATa ura3-52 pUDC100 (CEN6 ARS4 URA3 TEF1 _p - groEL-ACT1 _i , TPI1 _p -groES-PGI1 _i , cbmM-GroEL linker)	This study.
IMI229	MATa ura3-52 leu2-3, 112 can1A::GAL1 _p -prk-CYC1 _i , PGI1 _p -cbmQ2-TEF2 _i , PGK1 _p -cbmO2-ADH1 _i , TEF1 _p - groEL-ACT1 _i , TPI1 _p -groES-PGI1 _i , KILEU2	This study.
IMI232	MATa ura3-52 leu2-3, 112 can1A::KILEU2	This study.
IMU032	IMI232 p426_GPD (2μ URA3)	This study.
IMU033	IMI229 pUDC100 (CEN6 ARS4 URA3 TEF1 _p -groEL- ACT1 _i , TPI1 _p -groES-PGI1 _i , cbmM-GroEL linker)	This study.

The strain IMC014 that co-expressed the Rubisco form II ccbM and the four chaperones ccbQ2, ccbO2, groEL, and groES was constructed using in vivo transformation associated recombination. 200 fmol of each expression cassette were pooled with 100 fmol of the KpnI/SacI linearized pRS416 backbone in a final volume of 50 μl and transformed in CEN.PK 113-5D using the lithium acetate protocol (Gietz, et al., Yeast Transformation by the LiAc/SS Carrier DNA/PEG Method in Yeast Protocol, *Humana press*, 2006). Cells were selected on synthetic medium. Correct assembly of the fragment of pUDC075 was performed by multiplex PCR on transformant colonies using primers enabling amplification over the regions used for homologous recombination (Table 5) and by restriction analysis after re-transformation of the isolated plasmid in *E. coli* DH5a. pUDC075 was sequenced by Next-Generation Sequencing (Illumina, San Diego, California, U.S.A.) (100bp reads paired-end, 50 Mb) and assembled with Velvet (Zerbino, et al., Velvet: Algorithms for De Novo Short Read Assembly Using De Bruijn Graphs, *Genome Research*, 2008). The assembled sequence did not contain mutations in any of the assembled expression cassettes. The strains IMC034 and IMC035 that expressed ccbM/ccbQ2/ccbO2 and ccbM/groEL/groES respectively were constructed using the same in vivo assembly method with the following modification. To construct plasmids pUDC099 and pUDC100, 120 bp ccbO2-pRS416 linker and ccbM-GroEL linker were used to close the assembly respectively (Table 5), 100 fmol of each of complementary 120 bp oligonucleotides were added to the transformation. The strain IMC033 that only expressed the ccbM gene was constructed by transforming CEN.PK113-5D with pUDC098.

To construct the strain IMU033 that co-expressed PRK, ccbM, ccbQ2, ccbO2, GroEL, GroES, the intermediate strain IMI229 was constructed by integrating PRK, the four chaperones and KILEU2 (Guldener, et al., A second set of loxP marker cassettes for Cre-mediated multiple gene knockouts in budding yeast, *Nucleic Acids Research*, 2002) at the CAN1 locus by in vivo homologous integration in CEN.PK102-3A. The expression cassettes were PCR amplified using Phusion Hot-Start Polymerase (Finnzymes, Thermo Fisher Scientific Inc. Massachusetts, U.S.A.), the corresponding oligonucleotides and DNA templates (Table 5). Finally, the strain IMI229 was transformed with

pUDC100 that carries the Rubisco form II ccbM and the two *E. coli* chaperones groEL and groES.

Strain IMI232 was constructed by transforming CEN.PK102-3A with the KILEU2 cassette. IMI232 was finally transformed with the plasmid p426GPD to restore prototrophy resulting in the reference strain IMU032.

Example 3. Experimental Set-Up of Chemostat and Batch Experiments

Anaerobic chemostat cultivation was performed essentially as described (Basso, et al., Engineering topology and kinetics of sucrose metabolism in *Saccharomyces cerevisiae* for improved ethanol yield, *Metabolic Engineering* 13:694-703, 2011), but with 12.5 g l⁻¹ glucose and 12.5 g l⁻¹ galactose as the carbon source and where indicated, a mixture of 10% CO₂/90% N₂ replaced pure nitrogen as the sparging gas. Residual glucose and galactose concentrations were determined after rapid quenching (Mashego, et al., Critical evaluation of sampling techniques for residual glucose determination in carbon-limited chemostat culture of *Saccharomyces cerevisiae*, *Biotechnology and Bioengineering* 83:395-399, 2003) using commercial enzymatic assays for glucose (Boehringer, Mannheim, Germany) and D-galactose (Megazyme, Bray, Ireland). Anaerobic bioreactor batch cultures were grown essentially as described (Guadalupe Medina, et al., Elimination of glycerol production in anaerobic cultures of a *Saccharomyces cerevisiae* strain engineered to use acetic acid as an electron acceptor. *Applied and Environmental Microbiology* 76:190-195, 2010), but with 20 g L⁻¹ galactose and a sparging gas consisting of 10% CO₂ and 90% N₂. Biomass and metabolite concentrations in batch and chemostat and batch cultures were determined as described by Guadalupe et al. (Guadalupe Medina, et al., Elimination of glycerol production in anaerobic cultures of a *Saccharomyces cerevisiae* strain engineered to use acetic acid as an electron acceptor. *Appl. Environ. Microbiol.* 76, 190-195, 2010). In calculations of ethanol fluxes and yields, ethanol evaporation was corrected for based on a first-order evaporation rate constant of 0.008 h⁻¹ in the bioreactor set-ups and under the conditions used in this study.

Example 4. Enzyme Assays for Phosphoribulokinase (PRK) and Rubisco

Cell extracts for analysis of phosphoribulokinase (PRK) activity were prepared as described previously (Abbott, et

al., Catalase Overexpression reduces lactic acid-induced oxidative stress in *Saccharomyces cerevisiae*, *Applied and Environmental Microbiology* 75:2320-2325, 2009). PRK activity was measured at 30° C. by a coupled spectrophotometric assay (MacElroy, et al., Properties of Phosphoribulokinase from *Thiobacillus neapolitanus*, *Journal of Bacteriology* 112:532-538, 1972). Reaction rates were proportional to the amounts of cell extract added. Protein concentrations were determined by the Lowry method (Lowry, et al., Protein measurement with the Folin phenol reagent, *The Journal of Biological Chemistry* 193:265-275, 1951) using bovine serum albumin as a standard.

Cell extracts for Rubisco activity assays were prepared as described in Abbott, D. A. et al. Catalase overexpression reduces lactic acid-induced oxidative stress in *Saccharomyces cerevisiae*. *Appl. Environ. Microbiol.* 75:2320-2325, 2009, with two modifications: Tris-HCl (1 mM, pH 8.2) containing 20 mM MgCl₂·6H₂O, 5 mM of DTT 5 mM NaHCO₃ was used as sonication buffer and Tris-HCl (100 mM, pH 8.2), 20 mM MgCl₂·6H₂O and 5 mM of DTT as freezing buffer. Rubisco activity was determined by measuring ¹⁴CO₂-fixation (PerkinElmer, Groningen, The Netherlands) as described (Beudeker, et al., Relations between d-ribulose-1,5-biphosphate carboxylase, carboxysomes and CO₂ fixing capacity in the obligate chemolithotroph *Thiobacillus neapolitanus* grown under different limitations in the chemostat, *Archives of Microbiology* 124:185-189, 1980) and measuring radioactive counts in a TRI-CARB® 2700TR Series liquid scintillation counter (PerkinElmer, Groningen, The Netherlands), using Ultima Gold™ scintillation cocktail (PerkinElmer, Groningen, The Netherlands). Protein concentrations were determined by the Lowry method (Lowry, O. H., Rosebrough, N. J., Farr, A. L., & Randall, R. J. Protein measurement with the Folin phenol reagent. *J. Biol. Chem.* 193:265-275, 1951) using standard solutions of bovine serum albumin dissolved in 50 mM Tris-HCl (pH 8.2).

Example 5. The Activity of Rubisco and the Activity of PRK in Cell Extracts

In order to study a possible requirement of heterologous chaperones of Rubisco in *S. cerevisiae*, the form-II Rubisco-encoding cbbM gene from *T. denitrificans* was codon-optimised and expressed from a centromeric vector, both alone and in combination with expression cassettes for the codon-optimised *E. coli* groEL/groES and/or *T. denitrificans* cbbO2/cbbQ2 genes. Analysis of ribulose-1,5-biphosphate-dependent CO₂ fixation by yeast cell extracts demonstrated that functional expression of *T. denitrificans* Rubisco in *S. cerevisiae* was observed upon co-expression of *E. coli* GroEL/GroES. Rubisco activity increased from <0.2 nmol·min⁻¹·(mg protein)⁻¹ to more than 6 nmol·min⁻¹·(mg protein)⁻¹. Results of these experiments are visualised in FIG. 1, showing specific ribulose-1,5-bisphosphate carboxylase (Rubisco) activity in cell extracts of *S. cerevisiae* expressing Rubisco form II CbbM from *T. denitrificans* either alone (IMC033) or in combination with the *E. coli* chaperones GroEL/GroES (IMC035), The *T. denitrificans* chaperones CbbO2/CbbQ2 [20] (IMC034) or all four chaperones (IMC014). Heterologously expressed genes were codon optimised for expression in yeast and expressed from a single centromeric vector. Biomass samples were taken from anaerobic batch cultures on synthetic media (pH 5.0, 30° C.), sparged with nitrogen and containing 20 g l⁻¹ glucose as carbon source. Rubisco activities, measured as ¹⁴CO₂-fixation in cell extracts, in a wild-type reference

strain and in *S. cerevisiae* strains expressing cbbM and cbbM-cbbQ2-cbbO2 were below the detection limit of the enzyme assay (0.2 nmol CO₂ min⁻¹ mg protein⁻¹)

Co-expression of CbbO2/cbbQ2 did not result in a significant further increase of Rubisco activity. The positive effect of GroEL/GroES on Rubisco expression in *S. cerevisiae* demonstrates the potential value of this approach for metabolic engineering, especially when prokaryotic enzymes need to be functionally expressed in the cytosol of eukaryotes.

The Spinach *oleracea* PRK gene was integrated together with *E. coli* groEL/groES and *T. denitrificans* cbbO2/cbbQ2 into the *S. cerevisiae* genome at the CAN1 locus, under control of the galactose-inducible GAL1 promoter. This induced in high PRK activities in cell extracts of *S. cerevisiae* strain IMU033, which additionally carried the centromeric expression cassette for *T. denitrificans* Rubisco. This engineered yeast strain was used to quantitatively analyze the physiological impacts of the expression of Rubisco and PRK.

TABLE 8

	IMU032 (reference strain)		IMU033 (expressing PRK and Rubisco)	
CO ₂ in inlet gas (%)	0	10	0	10
CO ₂ in outlet gas (%)	0.89 ± 0.03	10.8 ± 0.0	1.02 ± 0.00	10.8 ± 0.1
Phosphoribulokinase (μmol mg protein ⁻¹ min ⁻¹)	0.58 ± 0.09	0.51 ± 0.12	14.4 ± 1.5	15.2 ± 1.0
Rubisco (nmol mg protein ⁻¹ min ⁻¹)	<0.2*	<0.2	4.59 ± 0.30	2.67 ± 0.28
Biomass yield on sugar (g g ⁻¹)	0.083 ± 0.000 ^a	0.084 ± 0.000 ^b	0.093 ± 0.001 ^a	0.095 ± 0.000 ^b
Ethanol yield on sugar (mol mol ⁻¹)	1.56 ± 0.03 ^c	1.56 ± 0.02 ^d	1.73 ± 0.02 ^c	1.73 ± 0.01 ^d
Glycerol yield on sugar (mol mol ⁻¹)	0.14 ± 0.00 ^e	0.12 ± 0.00 ^f	0.04 ± 0.00 ^{e,g}	0.01 ± 0.00 ^{f,g}

Table 8 show increased ethanol yields on sugar of an *S. cerevisiae* strain expressing phosphoribulokinase (PRK) and Rubisco. Physiological analysis of *S. cerevisiae* IMU033 expressing PRK and Rubisco and the isogenic reference strain IMU032 in anaerobic chemostat cultures, grown at a dilution rate of 0.05 h⁻¹ on a synthetic medium (pH 5) supplemented with 12.5 g l⁻¹ glucose and 12.5 g l⁻¹ galactose as carbon sources. To assess the impact of CO₂ concentration, chemostat cultures were run sparged either with pure nitrogen gas or with a blend of 10% CO₂ and 90% nitrogen. Results are represented as average ± mean deviations of data from independent duplicate chemostat experiments. Data pairs labelled with the same subscripts (a,a, b,b, etc.) are considered statistically different in a standard t-test (p<0.02).

Expression of Rubisco and the four chaperones without co-expression of PRK (strain IMC014) did not result in decreased glycerol yield (0.13 mol mol⁻¹) compared to the reference strain IMU032 (0.12 mol mol⁻¹) in carbon-limited chemostat cultures supplemented with CO₂, indicating that expression of a phosphoribulokinase (PRK) gene is required for the functional pathway in *S. cerevisiae* to decrease glycerol production. The physiological impact of expression of PRK and Rubisco on growth, substrate consumption and product formation in galactose-grown anaerobic batch cultures of *S. cerevisiae* was also investigated and compared with an isogenic reference strain. Growth conditions: T=30° C., pH 5.0, 10% CO₂ in inlet gas. Two independent replicate experiments were carried out, whose growth kinetic param-

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eters differed by less than 5%. Ethanol yield on galactose was 8% higher and glycerol production was reduced by 60% in the yeast cell in which PRK and Rubisco were functionally expressed, compared to the yeast cell lacking these enzymes. The differences were statistically significant (standard t-test (p value <0.02)). The activities of phosphoribu-

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lokinase and of Rubisco in cell extracts of the engineered strain IMU033 (table 7) enable the use of CO₂ as an electron acceptor. The ethanol yields and glycerol yields of strain IMU033 relative to the reference strain IMU032 (table 8) show that this is possible in an anaerobic fermentation with increased ethanol production.

SEQUENCES

Rubisco cbbM gene (synthetic; based on cbbM gene from *thiobacillus denitrificans*-pBTWW002, codon optimized Source: Hernandez et al 1996, GenBank ID: L37437.2)

SEQ ID NO: 1

ATGGATCAATCTGCAAGATATGCTGACTTGTCTTTAAAGGAAGAAGATTGATTAAGGTG
GTAGACATATTTTGGTTGCTTACAAAATGAAACCAAATCTGGTTATGGTTATTTGGAAGC
TGCTGCTCATTTTGGCTGCTGAATCTTCTACAGGTACAAATGTTGAAGTTTCTACTACAGATG
ATTTTACAAAAGGTGTTGATGCTTTAGTTTACTACATCGATGAAGCTTCAGAAGATATGAG
AATTGCTTATCCATTGGAATTATTCGACAGAAATGTTACTGACGGAAGATTTCATGTTAGTTT
CTTTTTGACTTTGGCTATTGGTAACAATCAAGGAATGGGAGATATAGAACATGCAAAAAT
GATAGATTTTACGTTCCAGAAAGATGTATTCAAATGTTTGTATGGTcCAGCTACAGATATTT
CTAATTTGTGGAGAATTTTGGGTAGACCAGTAGTTAATGGTGGTTATATTGCTGGTACTAT
TATTAAGCCAAAATTGGGTTTAAGACCAGAAcCATTGCTAAAGCTGCTTATCAATTTTGG
TTGGGTGGAGATTTTATCAAGAATGACGAACCACAAGGTAATCAAGTTTTTGTCCATTGA
AAAAAGTTTTGCCATTGGTTTACGATGCTATGAAAAGAGCACAGATGATACTGGTCAAGC
AAAATTGTTTTCTATGAATATTACTGCAGACGATCATTATGAAATGTGTCAAGAGCTGAT
TATGCTTTGGAAGTTTTCGGTCCAGATGCAGATAAATTGGCTTTTTTGGTAGATGGTTACG
TTGGAGGTCCAGGAATGGTTACTACTGCTAGAAGGCAATATCCTGGTCAATATTTGCATTA
TCATAGAGCAGGTACGGTGCTGTTACTTCTCCATCTGCTAAAAGAGGTATACTGCTTTT
GTTTTGGCTAAAATGTCTAGATTGCAAGGCGCTTCAGGTATTCATGTTGGTACTATGGGT
ATGGA AAAATGGAAGGAGAAGGCGACGATAAGATTATTGCTTATATGATGAAAAGGACG
AATGTCAAGGTCCAGTTTATTTCAAAAATGGTACGGTATGAAACCACTACTCCAATTAT
CTCCGGAGGAATGAATGCTTTGAGATTGCCTGGTTTTTTCGAAAATTTGGGTATGGTAAC
GTTATTAATACTGCAGGTGGTGGTTCTTACGGTCATATTGATTCTCTGCTGCTGGTGCTA
TTTCTTTGAGACAATCTTACGAATGTTGGAACAAGGTGCAGATCCAATTGAATTTGCTAA
GGAACATAAGGAATTTGCAAGAGCTTTTGAATCTTTTCCAAAAGATGCTGATAAGTTATTT
CCAGGATGGAGAGAAAATTTGGAGTTCATTCTTAA

Translated protein sequence of cbbM gene from *Thiobacillus denitrificans*

SEQ ID NO: 2

MDQSARYADLSLKEEDLIKGRHILVAYKMKPKSGYGYLEAAAHFAESSTGTNVEVSTTD
DFTKGVDAVYYIDEASEDMRIAYPLELFDRNVTDFRMLVVSFLTLAGNNGMGDIEHAK
MIDFYVPERCIQMPDGPATDISNLWRILGRPVVNGGYIAGTIIKPKLGLRPEPFAKAAAYQFWL
GGDFIKNDEPQGNQVFCPLKKVPLVVDAMKRAQDDTGQAKLFMSNITADHYEMCARAD
YALEVFGPDADKLAFLVDGYVGGPGMVTARRQYPGQYLHYHRAGHGAVTSPSAKRGYTA
FVLAKMSRLQGASGIHVGTMGYGKMEGEGDDKIIAYMIERDECQGPVYFQKWMKPTTP
IISGGMNALRLPGFFENLGHGNVINTAGGGSYGHIDSPAAGAISLRQSYECWKQGADPIEFA
KEHKEFARAPESFPKADADKLPFGWREKLGVHS

-continued

prk gene from *Spinacea oleracea*-pBTWW001, plasmid constructed using restriction and ligation. Source: Milanez and Mural 1988, GenBank ID: M21338.1

SEQ ID NO: 3

ATGTCACAACAACAACAATTGTGATTGGTTTAGCAGCAGATTGAGTTGTGGTAAGAGTA
CATTTCATGAGGAGGTTAACAAGTGTTCGCGTGGCGCGGCCGAGCCACCAAGGGTGGTA
ACCCAGATTCAAAACACATTGATTAGTGACACTACTACTGTTATCTGTTGGATGATTTTCAT
TCCCTTGATAGAAATGGCAGGAAAGTGGAAAAGTTACTGCTTTAGACCCAAAAGCTAATG
ATTTTGATCTTATGTATGAACAAGTTAAGGCTTTGAAAGAAGGTAAAGCTGTTGATAAACC
TATTTATAATCATGTTTCTGGTTTGTGGACCTCCTGAGCTTATTCAACCTCCTAAGATCT
TGGTTCATTGAAGGGTTACACCCCATGTATGACGCACGTGTGAGGGAATTGCTAGACTTCAG
CATCTACTTGGACATTAGCAATGAAGTTAAATTTGCCTGGAAAATTCAGAGAGACATGAAA
GAAAGAGGACACAGTCTTGAAAGCATCAAAGCCAGTATTGAATCCAGAAAGCCAGATTTTG
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ACTCATTCTGATGATGATGAAGGCAAAGTGTGAGAGTAAGGATGATTGAGAAAGAAGG
AGTCAAGTTTTTCAACCCAGTTTACTTGTGATGAAGGATCTACCATTTCATGGATTCCAT
GTGGTAGAAAATTAACATGTTCTTACCCTGGTATCAAATTTTCTATGGCCAGACACCTT
CTATGGCAACGAGGTGACAGTAGTAGAGATGGATGGGATGTTTGACAGATTAGACGAACT
AATCTACGTCGAAAGCCATTTGAGCAATCTATCAACCAAGTTTTATGGTGAAGTCACTCAA
CAAATGTTGAAGCACCAAAATTTCCAGGAAGCAACAATGGAACGTTTCTTCCAAACCA
TAATTGGATTGAAGATCAGAGACTTGTTCGAGCAGCTCGTTGCTAGCAGGTCTACAGCAAC
TGCAACAGCTGCTAAAGCCTAG

Translated protein sequence of prk gene from *Spinacea oleracea*

SEQ ID NO: 4

MSQQQTIVIGLAADSGCGKSTFMRRLTSVFGGAAEPPKGGNPDNLTISDTTVICLDDPHS
LDRNGRKVEKVTALDPKANDFDLMYEQVKALKEGKAVDKPIYNHVSGLLDPPELIQPPKIL
VIEGLHPMYDARVRELLDPSIYLDISNEVKFAWKIQRDMKERGHSLESIKASIESRKPDPDA
YIDPQKQHADVIVLELPTELIPDDDEGKVLVRMIQKEGVKFFNPVYLFDEGSTISWIPCGR
KLTCSPYGIKFSYGPDTFYGNEVTVMEDGMFDRLELIYVESHLNLSSTKIFYGEVTQQML
KHQNFPGSNNGTGFFQTIIGLKIRDLFEQLVASRSTATATAKA

SEQ ID NO: 5

cbbQ2 gene (synthetic, based on cbbQ2 gene from *Thiobacillus denitrificans*-codon optimized, original sequence obtained from Beller et al 2006, GenBank Gene ID: 3672366, Protein ID: AAZ98590.1

ATGACTACTAACAGGAACAATACAAGGTTACCAAGAACCATACTACCAAGCTCAAGGTA
GAGAAGTTCAATTGTACGAAGCTGCTTACAGAAACAGATTGCCAGTTATGGTTAAGGGTCC
AACTGGTTGTGTAAGTCTAGATTCGTTGAATACATGGCTTGAAGTTGAACAAGCCATTG
ATCACTGTTGCTTGTAAACGAAGACATGACTGCTTCTGACTTGGTTGGTAGATACTTGTGG
AAGCTAACGGTACTAGATGGTTGGACGGTCCATTGACTACTGCTGCTAGAATCGGTGCTAT
CTGTTACTTGGACGAAGTTGTTGAAGCTAGACAAGACACTACTGTTGTTATCCACCCATTG
ACTGACCACAGAAGAACTTTGCCATTGGACAAGAAGGGTGAATTGATCGAAGCTCACCCA
GACTTCCAATTGGTTATCTCTTACAACCCAGGTTACCAATCTTTGATGAAGGACTTGAAGC
AATCTACTAAGCAAAGATTGCTGCTTTCGACTTCGACTACCCAGACGCTGCTTGGAAAC
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GGTGGTGTGCTAGAAACTGAAGGGTCACGGTTTGGACGAAGGTATCTCTACTAGATTGT

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TGGTTTACGCTGCTACTTTGATGAAGGACGGTGTGACGCTGGTGACGCTTGTAGAATGGC
TTTGGTTAGACCAATCACTGACGACGCTGACATCAGAGAACTTTGGACCACGCTATCGAC
GCTACTTTTCGCTTAA

Translated protein sequence of cbbQ2 gene from *Thiobacillus
denitrificans*

SEQ ID NO: 6

MTTNKEQYKVHQEPYYQAQGREGVQLYEAAAYRNRLPVMVKGPTGCGKSRFVEYMAWKLNK
PLITVACNEDMTASDLVGRYLLEANGTRWLDGPLTTAARIGAICYLDEVVEARQDTTVVIHP
LTDHRRTLPLDKKGELIEAHPDFQLVISYNPGYQSLMKDLKQSTKQRFAPDFDYPDAALE
TTILARETGLDETTAGRLVKIGGVARNLKGHGLDEGISTRLLVYAATLMKDGVDAGDACRM
ALVRPITDDADIRETLDHAIDATFA

cbbQ2 gene (Synthetic, based on cbbQ2 gene from *Thiobacillus
denitrificans*-codon optimized, original sequence obtained from
Beller et al 2006, GenBank Gene ID: 3672365, Protein ID:
YP_316394.1

SEQ ID NO: 7

ATGGCTGCTTACTGGAAGGCTTTGGACACTAGATTGCTCAAGTTGAAGAAGTTTTCGACG
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TGGTAGAGTTATCGGTAAGTTGGGTAGAGGTGTTGAACCAATGTTGGCTTTCTTGAAGAA
TGCCATCTACTGCTCAAGCTGTTGGTGAAGCTGCTTTGCCAATGGTTATGGCTTTGATCC
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AGTTGCTAGAAGATTGCAATCTGCTGAACAATTGCAACACTACGTTGACGTTACTTTGGAC
TTCATGACTAGAACTACTGGTTCTATCCACGGTCACCACACTACTTTCCCATCTCCAGGTTT
GCCAGAATCTTCGCTCAAGCTCCAACTTGTGTAACCAATTGACTTTGGCTGGTTTGAGA
AACTGGGTGAATACGGTATCAGAACTACGGTACTCACCAGAAAGACAACAAGACTACT
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CCAGAATGGGACTACACTTCTCAAATTCAGACCAGACTGGGTTTCTGTTTACGAAGGTT
TGCACAGATCTGGTAACGCTGGTGACATCGACAGATTGTTGGCTAAGCACGCTGCTTTGGC
TAAGAGATTGAAGAAGATGTTGGACTTGTGAGCCACAAGACAAGGTTAGAGTTAGATAC
CAAGAAGAAGGTTCTGAATTGGACTTGGACGTTGCTATCAGATCTTGATCGACTTCAAGG
GTGGTGCTACTCCAGACCAAGAATCAACATGTCTCACAGATCTGACGGTAGAGACATCGC

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GTGACCCATTTCGCTATCGCTGGTTTCCACTCTAACACTAGACACGACGTTAGATACTTCCA
CATCAAGGGTTACTCTGAAAGATGGAACGACGACGTTAAGGCTAGATTGGCTGCTATGGA
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AGACCAGCTGACAAGAAGTTGATGTTGATCTTGACTGACGGTAGACCATCTGACGTTGACG
CTGCTGACGAAAGATTGTTGGTTGAAGACGCTAGACAAGCTGTTAAGGAATTGGACAGAC
AAGGTATCTTCGCTTACTGTATCTCTTTGGACGCTCAATTGAAGGCTGGTCTGACGACTA
CGTTGCTGAAATCTTCGCTAGACAATACACTGTTATCGACAGAGTTGAAAGATTGCCAGAA
AGATTGCCAGAATTGTTTCATGGCTTTGACTAAGTAA

Translated protein sequence of cbb02 gene from *Thiobacillus denitrificans*

SEQ ID NO: 8

MAAYWKALDTRFAQVEEVFDDCMAEALTVLSAEGVAAYLEAGRVIGKLRGVPEMLAFLE
EWPSTAQAVGEAALPMVMALIQRMQKSPNGKAIAPFLQTLAPVARRLQSAEQQLQHYVDVTL
DFMTRTTGSIHGHTTFFSPGLPEFFAQAPNLLNQLTLAGLRNWVEYGI RNYGTHPERQQD
YPSLQSDARAVLQREHGTLLVDVERKLDLYLRGLWQDHDHLVPYSTAFDEIRKVPYPYD
KLGMRLLPDVYDDLVLPCPAGRGAGGEDVLVSGLDRYRATLAHVMGHRRWSEAQIADNW
SPFQRMAVEFFEDCRVETLLMREYPGLARIFRALHPKPVEAACDGETTSCLRHRLAMLSRA
FIDPDHGYAAPVNLDFVARFPHARLADGTSSTSEMADLALSYAKTRRPSDQFAKVHFDDTV
VDYRDDNRQLWKFIIEEGDEEEAFDAKRKIEPEGEEIQGLPPRHYPEWDYTSQTYRPDWVSVY
EGLHRSNGAGDIDRLAKHAALAKRLKKMLDLLKPQDKVRVRYQEEGSELDLDVAIRSLID
FKGGATPDPRINMSHRSDGRDIAVMLLLDLSESLNEKAAGAGQTILELSQEAVSLLAWSIEK
LGDPFAIAGFHSNTRHDVRYFHIKGYSERWDDVKARLAAMEAGYSTRMGAMRHAHYL
SARPADKKMLLILTDGRPSDVDAADERLLVEDARQAVKELDRQGI FAYCISLDAQLKAGAD
DYVAEIFGRQYTVIDRVERLPERLPELFMALTK

GroEL gene (synthetic, based on GroEL from *E. coli*-codon optimized, original sequence obtained from Durfee et al 2008, Gene ID: 6061450, Protein ID: YP_001732912.1

SEQ ID NO: 9

ATGGCTGCTAAGGACGTTAAGTTCGGTAACGACGCTAGAGTTAAGATGTTGAGAGGTGTT
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ATTGGAAGACAAGTTCGAAAACATGGGTGCTCAAATGGTTAAGGAAGTTGCTTCTAAGGCT
AACGACGCTGCTGGTGACGGTACTACTACTGCTACTGTTTTGGCTCAAGCTATCATCACTG
AACGACGCTGCTGGTGACGGTACTACTACTGCTACTGTTTTGGCTCAAGCTATCATCACTG
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AGGCTGTACTGCTGCTGTTGAAGAATTGAAGGCTTTGTCTGTTCCATGTTCTGACTCTAA
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GCTGAAGCTATGGACAAGGTTGGTAAGGAAGGTGTTATCACTGTTGAAGACGGTACTGGT
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AACACTGTTAAGGGTGGTGACGGTAACTACGGTTACAACGCTGCTACTGAAGAATACGGT
AACATGATCGACATGGGTATCTTGGACCCAAGTAAGGTTACTAGATCTGCTTTGCAATAGG
CTGCTTCTGTTGCTGGTTTGATGATCACTACTGAATGTATGGTTACTGACTTGCCAAAGAA
CGACGCTGCTGACTTGGGTGCTGCTGGTGGTATGGTGGTATGGTGGTATGGTGGTAT
GATGTAA

Translated protein sequence of GroEL gene from *E. coli*
SEQ ID NO: 10
MAAKDVKFGNDARVKMLRGVNLADAVKVTLPKGRNVLDKSFAPITITKDGVSVAREI

ELEDKFENMGAQMVKVASKANDAAGDGTATVLAQAIITEGLKAVAAGMNPMDLKRGI
DKAVTAAVEELKALSVPSCSKAIAQVGTISANSDETVGKLIAEAMDKVGKEGVITVEDGTG
LQDEL DVVEGMQFDRGYLSPYFINKPETGAVELESPFILLADKKISNIREMLPVLEAVAKAG
KPLIIIAEDVEGEALATLVNMTMRGIVKVAAPGFGDRRKAMLQDIATLTGGTVISEEIG
MELEKATLEDLGQAKRVVINKDTTIIIDGVGEEAAIQGRVAQIRQQIEEATSDYDREKLQER
VAKLAGGVAVIKVGAATEVEMKEKKARVEDALHATRAAVEEGVVAGGGVALIRVASKLADL
RGQNEQNVGIIKVALRAMEAPLRQIVLNCGEPSVVANTVKGGDGNYGNAATEEYGNMI
DMGILDPTKVTRSALQYAASVAGLMITTECMVTDLPKNDAADLGAAGGMGGMGMM

GroES gene (synthetic, based on GroES *E. coli*-codon optimized,
original sequence obtained from Durfee et al 2008, Gene ID:
6061370, Protein ID: YP_001732911.1

SEQ ID NO: 11

ATGAACATCAGACCATTGCACGACAGATTATCGTTAAGAGAAAGGAAGTTGAACTAAGT
CTGCTGGTGGTATCGTTTTGACTGGTCTGCTGCTGCTAAGTCTACTAGAGGTGAAGTTTT
GGCTGTTGGTAACGGTAGAATCTTGAAAACGGTGAAGTTAAGCCATTGGACGTTAAGGT
TGGTGACATCGTTATCTTCAACGACGGTTACGGTGTAAAGTCTGAAAAGATCGACAACGAA
GAAGTTTTGATCATGTCTGAATCTGACATCTTGGCTATCGTTGAAGCTTAA

Translated protein sequence of GroES gene from *E. coli*
SEQ ID NO: 12
MNIRPLHDRVIVKRKEVETKSAGGIVLTGSAAAKSTRGEVLAVGNRILENGEVKPLDVKV

GDIVIFNDGYGVKSEKIDNEEVLMSES DILAIVEA

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 45

<210> SEQ ID NO 1
<211> LENGTH: 1380
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic; based on cbbM gene from Thiobacillus
denitrificans - pBTWW002, codon optimized
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(1380)

<400> SEQUENCE: 1

atg gat caa tct gca aga tat gct gac ttg tct tta aag gaa gaa gat      48
Met Asp Gln Ser Ala Arg Tyr Ala Asp Leu Ser Leu Lys Glu Glu Asp
1          5          10          15

ttg att aaa ggt ggt aga cat att ttg gtt gct tac aaa atg aaa cca      96
Leu Ile Lys Gly Gly Arg His Ile Leu Val Ala Tyr Lys Met Lys Pro
          20          25          30

aaa tct ggt tat ggt tat ttg gaa gct gct gct cat ttt gct gct gaa     144
Lys Ser Gly Tyr Gly Tyr Leu Glu Ala Ala Ala His Phe Ala Ala Glu
          35          40          45

tct tct aca ggt aca aat gtt gaa gtt tct act aca gat gat ttt aca     192
Ser Ser Thr Gly Thr Asn Val Glu Val Ser Thr Thr Asp Asp Phe Thr
          50          55          60

aaa ggt gtt gat gct tta gtt tac tac atc gat gaa gct tca gaa gat     240
Lys Gly Val Asp Ala Leu Val Tyr Tyr Ile Asp Glu Ala Ser Glu Asp
        65          70          75          80

atg aga att gct tat cca ttg gaa tta ttc gac aga aat gtt act gac     288
Met Arg Ile Ala Tyr Pro Leu Glu Leu Phe Asp Arg Asn Val Thr Asp
          85          90          95

gga aga ttc atg tta gtt tct ttt ttg act ttg gct att ggt aac aat     336
Gly Arg Phe Met Leu Val Ser Phe Leu Thr Leu Ala Ile Gly Asn Asn
          100          105          110

caa gga atg gga gat ata gaa cat gca aaa atg ata gat ttt tac gtt     384
Gln Gly Met Gly Asp Ile Glu His Ala Lys Met Ile Asp Phe Tyr Val
          115          120          125

cca gaa aga tgt att caa atg ttt gat ggt cca gct aca gat att tct     432
Pro Glu Arg Cys Ile Gln Met Phe Asp Gly Pro Ala Thr Asp Ile Ser
          130          135          140

aat ttg tgg aga att ttg ggt aga cca gta gtt aat ggt ggt tat att     480
Asn Leu Trp Arg Ile Leu Gly Arg Pro Val Val Asn Gly Gly Tyr Ile
          145          150          155          160

gct ggt act att att aag cca aaa ttg ggt tta aga cca gaa cca ttt     528
Ala Gly Thr Ile Ile Lys Pro Lys Leu Gly Leu Arg Pro Glu Pro Phe
          165          170          175

gct aaa gct gct tat caa ttt tgg ttg ggt gga gat ttt atc aag aat     576
Ala Lys Ala Ala Tyr Gln Phe Trp Leu Gly Gly Asp Phe Ile Lys Asn
          180          185          190

gac gaa cca caa ggt aat caa gtt ttt tgt cca ttg aaa aaa gtt ttg     624
Asp Glu Pro Gln Gly Asn Gln Val Phe Cys Pro Leu Lys Lys Val Leu
          195          200          205

cca ttg gtt tac gat gct atg aaa aga gca caa gat gat act ggt caa     672
Pro Leu Val Tyr Asp Ala Met Lys Arg Ala Gln Asp Asp Thr Gly Gln
          210          215          220

gca aaa ttg ttt tct atg aat att act gca gac gat cat tat gaa atg     720
Ala Lys Leu Phe Ser Met Asn Ile Thr Ala Asp Asp His Tyr Glu Met
          225          230          235          240

tgt gca aga gct gat tat gct ttg gaa gtt ttc ggt cca gat gca gat     768
Cys Ala Arg Ala Asp Tyr Ala Leu Glu Val Phe Gly Pro Asp Ala Asp
          245          250          255

aaa ttg gct ttt ttg gta gat ggt tac gtt gga ggt cca gga atg gtt     816
Lys Leu Ala Phe Leu Val Asp Gly Tyr Val Gly Gly Pro Gly Met Val
          260          265          270

act act gct aga agg caa tat cct ggt caa tat ttg cat tat cat aga     864

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Thr	Thr	Ala	Arg	Arg	Gln	Tyr	Pro	Gly	Gln	Tyr	Leu	His	Tyr	His	Arg		
		275					280					285					
gca	ggt	cac	ggt	gct	gtt	act	tct	cca	tct	gct	aaa	aga	ggt	tat	act	912	
Ala	Gly	His	Gly	Ala	Val	Thr	Ser	Pro	Ser	Ala	Lys	Arg	Gly	Tyr	Thr		
		290				295					300						
gct	ttt	ggt	ttg	gct	aaa	atg	tct	aga	ttg	caa	ggc	gct	tca	ggt	att	960	
Ala	Phe	Val	Leu	Ala	Lys	Met	Ser	Arg	Leu	Gln	Gly	Ala	Ser	Gly	Ile		
305					310					315					320		
cat	ggt	ggt	act	atg	ggt	tat	gga	aaa	atg	gaa	gga	gaa	ggc	gac	gat	1008	
His	Val	Gly	Thr	Met	Gly	Tyr	Gly	Lys	Met	Glu	Gly	Glu	Gly	Asp	Asp		
				325					330					335			
aag	att	att	gct	tat	atg	ata	gaa	agg	gac	gaa	tgt	caa	ggt	cca	ggt	1056	
Lys	Ile	Ile	Ala	Tyr	Met	Ile	Glu	Arg	Asp	Glu	Cys	Gln	Gly	Pro	Val		
			340					345					350				
tat	ttt	caa	aaa	tgg	tac	ggt	atg	aaa	cca	act	act	cca	att	atc	ttc	1104	
Tyr	Phe	Gln	Lys	Trp	Tyr	Gly	Met	Lys	Pro	Thr	Thr	Pro	Ile	Ile	Ser		
		355				360						365					
gga	gga	atg	aat	gct	ttg	aga	ttg	cct	ggt	ttt	ttc	gaa	aat	ttg	ggt	1152	
Gly	Gly	Met	Asn	Ala	Leu	Arg	Leu	Pro	Gly	Phe	Phe	Glu	Asn	Leu	Gly		
		370			375						380						
cat	ggt	aac	ggt	att	aat	act	gca	ggt	ggt	ggt	tct	tac	ggt	cat	att	1200	
His	Gly	Asn	Val	Ile	Asn	Thr	Ala	Gly	Gly	Gly	Ser	Tyr	Gly	His	Ile		
					390					395					400		
gat	tct	cct	gct	gct	ggt	gct	att	tct	ttg	aga	caa	tct	tac	gaa	tgt	1248	
Asp	Ser	Pro	Ala	Ala	Gly	Ala	Ile	Ser	Leu	Arg	Gln	Ser	Tyr	Glu	Cys		
			405					410						415			
tgg	aaa	caa	ggt	gca	gat	cca	att	gaa	ttt	gct	aag	gaa	cat	aag	gaa	1296	
Trp	Lys	Gln	Gly	Ala	Asp	Pro	Ile	Glu	Phe	Ala	Lys	Glu	His	Lys	Glu		
			420					425				430					
ttt	gca	aga	gct	ttt	gaa	tct	ttt	cca	aaa	gat	gct	gat	aag	tta	ttt	1344	
Phe	Ala	Arg	Ala	Phe	Glu	Ser	Phe	Pro	Lys	Asp	Ala	Asp	Lys	Leu	Phe		
		435				440					445						
cca	gga	tgg	aga	gaa	aaa	ttg	gga	ggt	cat	tct	taa					1380	
Pro	Gly	Trp	Arg	Glu	Lys	Leu	Gly	Val	His	Ser							
		450				455											

<210> SEQ ID NO 2

<211> LENGTH: 459

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 2

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Leu	Ile	Lys	Gly	Gly	Arg	His	Ile	Leu	Val	Ala	Tyr	Lys	Met	Lys	Pro
		20					25					30			

Lys	Ser	Gly	Tyr	Gly	Tyr	Leu	Glu	Ala	Ala	Ala	His	Phe	Ala	Ala	Glu
		35				40					45				

Ser	Ser	Thr	Gly	Thr	Asn	Val	Glu	Val	Ser	Thr	Thr	Asp	Asp	Phe	Thr
		50			55					60					

Lys	Gly	Val	Asp	Ala	Leu	Val	Tyr	Tyr	Ile	Asp	Glu	Ala	Ser	Glu	Asp
65				70					75					80	

Met	Arg	Ile	Ala	Tyr	Pro	Leu	Glu	Leu	Phe	Asp	Arg	Asn	Val	Thr	Asp
			85					90					95		

Gly	Arg	Phe	Met	Leu	Val	Ser	Phe	Leu	Thr	Leu	Ala	Ile	Gly	Asn	Asn
		100					105					110			

Gln	Gly	Met	Gly	Asp	Ile	Glu	His	Ala	Lys	Met	Ile	Asp	Phe	Tyr	Val
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

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115	120	125
Pro Glu Arg Cys Ile Gln Met Phe Asp Gly Pro Ala Thr Asp Ile Ser		
130	135	140
Asn Leu Trp Arg Ile Leu Gly Arg Pro Val Val Asn Gly Gly Tyr Ile		
145	150	155
Ala Gly Thr Ile Ile Lys Pro Lys Leu Gly Leu Arg Pro Glu Pro Phe		
165	170	175
Ala Lys Ala Ala Tyr Gln Phe Trp Leu Gly Gly Asp Phe Ile Lys Asn		
180	185	190
Asp Glu Pro Gln Gly Asn Gln Val Phe Cys Pro Leu Lys Lys Val Leu		
195	200	205
Pro Leu Val Tyr Asp Ala Met Lys Arg Ala Gln Asp Asp Thr Gly Gln		
210	215	220
Ala Lys Leu Phe Ser Met Asn Ile Thr Ala Asp Asp His Tyr Glu Met		
225	230	235
Cys Ala Arg Ala Asp Tyr Ala Leu Glu Val Phe Gly Pro Asp Ala Asp		
245	250	255
Lys Leu Ala Phe Leu Val Asp Gly Tyr Val Gly Gly Pro Gly Met Val		
260	265	270
Thr Thr Ala Arg Arg Gln Tyr Pro Gly Gln Tyr Leu His Tyr His Arg		
275	280	285
Ala Gly His Gly Ala Val Thr Ser Pro Ser Ala Lys Arg Gly Tyr Thr		
290	295	300
Ala Phe Val Leu Ala Lys Met Ser Arg Leu Gln Gly Ala Ser Gly Ile		
305	310	315
His Val Gly Thr Met Gly Tyr Gly Lys Met Glu Gly Glu Gly Asp Asp		
325	330	335
Lys Ile Ile Ala Tyr Met Ile Glu Arg Asp Glu Cys Gln Gly Pro Val		
340	345	350
Tyr Phe Gln Lys Trp Tyr Gly Met Lys Pro Thr Thr Pro Ile Ile Ser		
355	360	365
Gly Gly Met Asn Ala Leu Arg Leu Pro Gly Phe Phe Glu Asn Leu Gly		
370	375	380
His Gly Asn Val Ile Asn Thr Ala Gly Gly Gly Ser Tyr Gly His Ile		
385	390	395
Asp Ser Pro Ala Ala Gly Ala Ile Ser Leu Arg Gln Ser Tyr Glu Cys		
405	410	415
Trp Lys Gln Gly Ala Asp Pro Ile Glu Phe Ala Lys Glu His Lys Glu		
420	425	430
Phe Ala Arg Ala Phe Glu Ser Phe Pro Lys Asp Ala Asp Lys Leu Phe		
435	440	445
Pro Gly Trp Arg Glu Lys Leu Gly Val His Ser		
450	455	

<210> SEQ ID NO 3

<211> LENGTH: 1059

<212> TYPE: DNA

<213> ORGANISM: Spinacia oleracea

<220> FEATURE:

<221> NAME/KEY: CDS

<222> LOCATION: (1)..(1059)

<400> SEQUENCE: 3

atg tca caa caa caa aca att gtg att ggt tta gca gca gat tca ggt
Met Ser Gln Gln Thr Ile Val Ile Gly Leu Ala Ala Asp Ser Gly
1 5 10 15

48

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tgt ggt aag agt aca ttc atg agg agg tta aca agt gtt ttc ggt ggc	96
Cys Gly Lys Ser Thr Phe Met Arg Arg Leu Thr Ser Val Phe Gly Gly	
20 25 30	
gcg gcc gag cca cca aag ggt ggt aac cca gat tca aac aca ttg att	144
Ala Ala Glu Pro Pro Lys Gly Gly Asn Pro Asp Ser Asn Thr Leu Ile	
35 40 45	
agt gac act act act gtt atc tgt ttg gat gat ttt cat tcc ctt gat	192
Ser Asp Thr Thr Thr Val Ile Cys Leu Asp Asp Phe His Ser Leu Asp	
50 55 60	
aga aat ggc agg aaa gtg gaa aaa gtt act gct tta gac cca aaa gct	240
Arg Asn Gly Arg Lys Val Glu Lys Val Thr Ala Leu Asp Pro Lys Ala	
65 70 75 80	
aat gat ttt gat ctt atg tat gaa caa gtt aag gct ttg aaa gaa ggt	288
Asn Asp Phe Asp Leu Met Tyr Glu Gln Val Lys Ala Leu Lys Glu Gly	
85 90 95	
aaa gct gtt gat aaa cct att tat aat cat gtt tct ggt ttg ttg gac	336
Lys Ala Val Asp Lys Pro Ile Tyr Asn His Val Ser Gly Leu Leu Asp	
100 105 110	
cct cct gag ctt att caa cct cct aag atc ttg gtc att gaa ggg tta	384
Pro Pro Glu Leu Ile Gln Pro Pro Lys Ile Leu Val Ile Glu Gly Leu	
115 120 125	
cac ccc atg tat gac gca cgt gtg agg gaa ttg cta gac ttc agc atc	432
His Pro Met Tyr Asp Ala Arg Val Arg Glu Leu Leu Asp Phe Ser Ile	
130 135 140	
tac ttg gac att agc aat gaa gtt aaa ttt gcc tgg aaa att cag aga	480
Tyr Leu Asp Ile Ser Asn Glu Val Lys Phe Ala Trp Lys Ile Gln Arg	
145 150 155 160	
gac atg aaa gaa aga gga cac agt ctt gaa agc atc aaa gcc agt att	528
Asp Met Lys Glu Arg Gly His Ser Leu Glu Ser Ile Lys Ala Ser Ile	
165 170 175	
gaa tcc aga aag cca gat ttt gat gct tac att gac cca caa aag cag	576
Glu Ser Arg Lys Pro Asp Phe Asp Ala Tyr Ile Asp Pro Gln Lys Gln	
180 185 190	
cat gct gat gta gtg att gaa gta ttg cca act gaa ctc att cct gat	624
His Ala Asp Val Val Ile Glu Val Leu Pro Thr Glu Leu Ile Pro Asp	
195 200 205	
gat gat gaa ggc aaa gtg ttg aga gta agg atg att cag aaa gaa gga	672
Asp Asp Glu Gly Lys Val Leu Arg Val Arg Met Ile Gln Lys Glu Gly	
210 215 220	
gtc aag ttt ttc aac cca gtt tac ttg ttt gat gaa gga tct acc att	720
Val Lys Phe Phe Asn Pro Val Tyr Leu Phe Asp Glu Gly Ser Thr Ile	
225 230 235 240	
tca tgg att cca tgt ggt aga aaa tta aca tgt tct tac cct ggt atc	768
Ser Trp Ile Pro Cys Gly Arg Lys Leu Thr Cys Ser Tyr Pro Gly Ile	
245 250 255	
aaa ttt tcc tat ggc cca gac acc ttc tat ggc aac gag gtg aca gta	816
Lys Phe Ser Tyr Gly Pro Asp Thr Phe Tyr Gly Asn Glu Val Thr Val	
260 265 270	
gta gag atg gat ggg atg ttt gac aga tta gac gaa cta atc tac gtc	864
Val Glu Met Asp Gly Met Phe Asp Arg Leu Asp Glu Leu Ile Tyr Val	
275 280 285	
gaa agc cat ttg agc aat cta tca acc aag ttt tat ggt gaa gtc act	912
Glu Ser His Leu Ser Asn Leu Ser Thr Lys Phe Tyr Gly Glu Val Thr	
290 295 300	
caa caa atg ttg aag cac caa aat ttc cca gga agc aac aat gga act	960
Gln Gln Met Leu Lys His Gln Asn Phe Pro Gly Ser Asn Asn Gly Thr	
305 310 315 320	
ggg ttc ttc caa acc ata att gga ttg aag atc aga gac ttg ttc gag	1008
Gly Phe Phe Gln Thr Ile Ile Gly Leu Lys Ile Arg Asp Leu Phe Glu	

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325	330	335	
cag ctc gtt gct agc agg tct aca gca act gca aca gct gct aaa gcc			1056
Gln Leu Val Ala Ser Arg Ser Thr Ala Thr Ala Thr Ala Ala Lys Ala			
340	345	350	
tag			1059
<210> SEQ ID NO 4			
<211> LENGTH: 352			
<212> TYPE: PRT			
<213> ORGANISM: Spinacia oleracea			
<400> SEQUENCE: 4			
Met Ser Gln Gln Gln Thr Ile Val Ile Gly Leu Ala Ala Asp Ser Gly			
1	5	10	15
Cys Gly Lys Ser Thr Phe Met Arg Arg Leu Thr Ser Val Phe Gly Gly			
	20	25	30
Ala Ala Glu Pro Pro Lys Gly Gly Asn Pro Asp Ser Asn Thr Leu Ile			
	35	40	45
Ser Asp Thr Thr Thr Val Ile Cys Leu Asp Asp Phe His Ser Leu Asp			
	50	55	60
Arg Asn Gly Arg Lys Val Glu Lys Val Thr Ala Leu Asp Pro Lys Ala			
	65	70	75
Asn Asp Phe Asp Leu Met Tyr Glu Gln Val Lys Ala Leu Lys Glu Gly			
	85	90	95
Lys Ala Val Asp Lys Pro Ile Tyr Asn His Val Ser Gly Leu Leu Asp			
	100	105	110
Pro Pro Glu Leu Ile Gln Pro Pro Lys Ile Leu Val Ile Glu Gly Leu			
	115	120	125
His Pro Met Tyr Asp Ala Arg Val Arg Glu Leu Leu Asp Phe Ser Ile			
	130	135	140
Tyr Leu Asp Ile Ser Asn Glu Val Lys Phe Ala Trp Lys Ile Gln Arg			
	145	150	155
Asp Met Lys Glu Arg Gly His Ser Leu Glu Ser Ile Lys Ala Ser Ile			
	165	170	175
Glu Ser Arg Lys Pro Asp Phe Asp Ala Tyr Ile Asp Pro Gln Lys Gln			
	180	185	190
His Ala Asp Val Val Ile Glu Val Leu Pro Thr Glu Leu Ile Pro Asp			
	195	200	205
Asp Asp Glu Gly Lys Val Leu Arg Val Arg Met Ile Gln Lys Glu Gly			
	210	215	220
Val Lys Phe Phe Asn Pro Val Tyr Leu Phe Asp Glu Gly Ser Thr Ile			
	225	230	235
Ser Trp Ile Pro Cys Gly Arg Lys Leu Thr Cys Ser Tyr Pro Gly Ile			
	245	250	255
Lys Phe Ser Tyr Gly Pro Asp Thr Phe Tyr Gly Asn Glu Val Thr Val			
	260	265	270
Val Glu Met Asp Gly Met Phe Asp Arg Leu Asp Glu Leu Ile Tyr Val			
	275	280	285
Glu Ser His Leu Ser Asn Leu Ser Thr Lys Phe Tyr Gly Glu Val Thr			
	290	295	300
Gln Gln Met Leu Lys His Gln Asn Phe Pro Gly Ser Asn Asn Gly Thr			
	305	310	315
Gly Phe Phe Gln Thr Ile Ile Gly Leu Lys Ile Arg Asp Leu Phe Glu			
	325	330	335

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Gln Leu Val Ala Ser Arg Ser Thr Ala Thr Ala Thr Ala Ala Lys Ala
 340 345 350

<210> SEQ ID NO 5
 <211> LENGTH: 807
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic, based on cbbQ2 gene from
 Thiobacillus denitrificans -codon optimized
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(807)

<400> SEQUENCE: 5

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Met Thr Thr Asn Lys Glu Gln Tyr Lys Val His Gln Glu Pro Tyr Tyr	
1 5 10 15	
caa gct caa ggt aga gaa gtt caa ttg tac gaa gct gct tac aga aac	96
Gln Ala Gln Gly Arg Glu Val Gln Leu Tyr Glu Ala Ala Tyr Arg Asn	
20 25 30	
aga ttg cca gtt atg gtt aag ggt cca act ggt tgt ggt aag tct aga	144
Arg Leu Pro Val Met Val Lys Gly Pro Thr Gly Cys Gly Lys Ser Arg	
35 40 45	
ttc gtt gaa tac atg gct tgg aag ttg aac aag cca ttg atc act gtt	192
Phe Val Glu Tyr Met Ala Trp Lys Leu Asn Lys Pro Leu Ile Thr Val	
50 55 60	
gct tgt aac gaa gac atg act gct tct gac ttg gtt ggt aga tac ttg	240
Ala Cys Asn Glu Asp Met Thr Ala Ser Asp Leu Val Gly Arg Tyr Leu	
65 70 75 80	
ttg gaa gct aac ggt act aga tgg ttg gac ggt cca ttg act act gct	288
Leu Glu Ala Asn Gly Thr Arg Trp Leu Asp Gly Pro Leu Thr Thr Ala	
85 90 95	
gct aga atc ggt gct atc tgt tac ttg gac gaa gtt gtt gaa gct aga	336
Ala Arg Ile Gly Ala Ile Cys Tyr Leu Asp Glu Val Val Glu Ala Arg	
100 105 110	
caa gac act act gtt gtt atc cac cca ttg act gac cac aga aga act	384
Gln Asp Thr Thr Val Val Ile His Pro Leu Thr Asp His Arg Arg Thr	
115 120 125	
ttg cca ttg gac aag aag ggt gaa ttg atc gaa gct cac cca gac ttc	432
Leu Pro Leu Asp Lys Lys Gly Glu Leu Ile Glu Ala His Pro Asp Phe	
130 135 140	
caa ttg gtt atc tct tac aac cca ggt tac caa tct ttg atg aag gac	480
Gln Leu Val Ile Ser Tyr Asn Pro Gly Tyr Gln Ser Leu Met Lys Asp	
145 150 155 160	
ttg aag caa tct act aag caa aga ttc gct gct ttc gac ttc gac tac	528
Leu Lys Gln Ser Thr Lys Gln Arg Phe Ala Ala Phe Asp Phe Asp Tyr	
165 170 175	
cca gac gct gct ttg gaa act act atc ttg gct aga gaa act ggt ttg	576
Pro Asp Ala Ala Leu Glu Thr Thr Ile Leu Ala Arg Glu Thr Gly Leu	
180 185 190	
gac gaa act act gct ggt aga ttg gtt aag atc ggt ggt gtt gct aga	624
Asp Glu Thr Thr Ala Gly Arg Leu Val Lys Ile Gly Gly Val Ala Arg	
195 200 205	
aac ttg aag ggt cac ggt ttg gac gaa ggt atc tct act aga ttg ttg	672
Asn Leu Lys Gly His Gly Leu Asp Glu Gly Ile Ser Thr Arg Leu Leu	
210 215 220	
gtt tac gct gct act ttg atg aag gac ggt gtt gac gct ggt gac gct	720
Val Tyr Ala Ala Thr Leu Met Lys Asp Gly Val Asp Ala Gly Asp Ala	
225 230 235 240	
tgt aga atg gct ttg gtt aga cca atc act gac gac gct gac atc aga	768
Cys Arg Met Ala Leu Val Arg Pro Ile Thr Asp Asp Ala Asp Ile Arg	

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245	250	255	
gaa act ttg gac cac gct atc gac gct act ttc gct taa			
Glu Thr Leu Asp His Ala Ile Asp Ala Thr Phe Ala			807
260	265		

<210> SEQ ID NO 6
 <211> LENGTH: 268
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 6

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Gln Ala Gln Gly Arg Glu Val Gln Leu Tyr Glu Ala Ala Tyr Arg Asn	
20 25 30	
Arg Leu Pro Val Met Val Lys Gly Pro Thr Gly Cys Gly Lys Ser Arg	
35 40 45	
Phe Val Glu Tyr Met Ala Trp Lys Leu Asn Lys Pro Leu Ile Thr Val	
50 55 60	
Ala Cys Asn Glu Asp Met Thr Ala Ser Asp Leu Val Gly Arg Tyr Leu	
65 70 75 80	
Leu Glu Ala Asn Gly Thr Arg Trp Leu Asp Gly Pro Leu Thr Thr Ala	
85 90 95	
Ala Arg Ile Gly Ala Ile Cys Tyr Leu Asp Glu Val Val Glu Ala Arg	
100 105 110	
Gln Asp Thr Thr Val Val Ile His Pro Leu Thr Asp His Arg Arg Thr	
115 120 125	
Leu Pro Leu Asp Lys Lys Gly Glu Leu Ile Glu Ala His Pro Asp Phe	
130 135 140	
Gln Leu Val Ile Ser Tyr Asn Pro Gly Tyr Gln Ser Leu Met Lys Asp	
145 150 155 160	
Leu Lys Gln Ser Thr Lys Gln Arg Phe Ala Ala Phe Asp Phe Asp Tyr	
165 170 175	
Pro Asp Ala Ala Leu Glu Thr Thr Ile Leu Ala Arg Glu Thr Gly Leu	
180 185 190	
Asp Glu Thr Thr Ala Gly Arg Leu Val Lys Ile Gly Gly Val Ala Arg	
195 200 205	
Asn Leu Lys Gly His Gly Leu Asp Glu Gly Ile Ser Thr Arg Leu Leu	
210 215 220	
Val Tyr Ala Ala Thr Leu Met Lys Asp Gly Val Asp Ala Gly Asp Ala	
225 230 235 240	
Cys Arg Met Ala Leu Val Arg Pro Ile Thr Asp Asp Ala Asp Ile Arg	
245 250 255	
Glu Thr Leu Asp His Ala Ile Asp Ala Thr Phe Ala	
260 265	

<210> SEQ ID NO 7
 <211> LENGTH: 2289
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic, based on cbb02 gene from
 Thiobacillus denitrificans -codon optimized
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2289)

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<400> SEQUENCE: 7

atg gct gct tac tgg aag gct ttg gac act aga ttc gct caa gtt gaa	48
Met Ala Ala Tyr Trp Lys Ala Leu Asp Thr Arg Phe Ala Gln Val Glu	
1 5 10 15	
gaa gtt ttc gac gac tgt atg gct gaa gct ttg act gtt ttg tct gct	96
Glu Val Phe Asp Asp Cys Met Ala Glu Ala Leu Thr Val Leu Ser Ala	
20 25 30	
gaa ggt gtt gct gct tac ttg gaa gct ggt aga gtt atc ggt aag ttg	144
Glu Gly Val Ala Ala Tyr Leu Glu Ala Gly Arg Val Ile Gly Lys Leu	
35 40 45	
ggt aga ggt gtt gaa cca atg ttg gct ttc ttg gaa gaa tgg cca tct	192
Gly Arg Gly Val Glu Pro Met Leu Ala Phe Leu Glu Glu Trp Pro Ser	
50 55 60	
act gct caa gct gtt ggt gaa gct gct ttg cca atg gtt atg gct ttg	240
Thr Ala Gln Ala Val Gly Glu Ala Ala Leu Pro Met Val Met Ala Leu	
65 70 75 80	
atc caa aga atg caa aag tct cca aac ggt aag gct atc gct cca ttc	288
Ile Gln Arg Met Gln Lys Ser Pro Asn Gly Lys Ala Ile Ala Pro Phe	
85 90 95	
ttg caa act ttg gct cca gtt gct aga aga ttg caa tct gct gaa caa	336
Leu Gln Thr Leu Ala Pro Val Ala Arg Arg Leu Gln Ser Ala Glu Gln	
100 105 110	
ttg caa cac tac gtt gac gtt act ttg gac ttc atg act aga act act	384
Leu Gln His Tyr Val Asp Val Thr Leu Asp Phe Met Thr Arg Thr Thr	
115 120 125	
ggt tct atc cac ggt cac cac act act ttc cca tct cca ggt ttg cca	432
Gly Ser Ile His Gly His His Thr Thr Phe Pro Ser Pro Gly Leu Pro	
130 135 140	
gaa ttc ttc gct caa gct cca aac ttg ttg aac caa ttg act ttg gct	480
Glu Phe Phe Ala Gln Ala Pro Asn Leu Leu Asn Gln Leu Thr Leu Ala	
145 150 155 160	
ggt ttg aga aac tgg gtt gaa tac ggt atc aga aac tac ggt act cac	528
Gly Leu Arg Asn Trp Val Glu Tyr Gly Ile Arg Asn Tyr Gly Thr His	
165 170 175	
cca gaa aga caa caa gac tac ttc tct ttg caa tct gct gac gct aga	576
Pro Glu Arg Gln Gln Asp Tyr Phe Ser Leu Gln Ser Ala Asp Ala Arg	
180 185 190	
gct gtt ttg caa aga gaa aga cac ggt act ttg ttg gtt gac gtt gaa	624
Ala Val Leu Gln Arg Glu Arg His Gly Thr Leu Leu Val Asp Val Glu	
195 200 205	
aga aag ttg gac ttg tac ttg aga ggt ttg tgg caa gac cac gac cac	672
Arg Lys Leu Asp Leu Tyr Leu Arg Gly Leu Trp Gln Asp His Asp His	
210 215 220	
ttg gtt cca tac tct act gct ttc gac gaa atc aga aag cca gtt cca	720
Leu Val Pro Tyr Ser Thr Ala Phe Asp Glu Ile Arg Lys Pro Val Pro	
225 230 235 240	
tac tac gac aag ttg ggt atg aga ttg cca gac gtt tac gac gac ttg	768
Tyr Tyr Asp Lys Leu Gly Met Arg Leu Pro Asp Val Tyr Asp Asp Leu	
245 250 255	
ggt ttg cca tgt cca gct ggt aga ggt ggt gct ggt ggt gaa gac gtt	816
Val Leu Pro Cys Pro Ala Gly Arg Gly Gly Ala Gly Gly Glu Asp Val	
260 265 270	
ttg gtt tct ggt ttg gac aga tac aga gct act ttg gct cac atg gtt	864
Leu Val Ser Gly Leu Asp Arg Tyr Arg Ala Thr Leu Ala His Met Val	
275 280 285	
ggt cac aga aga tgg tct gaa gct caa atc gct gac aac tgg tct cca	912
Gly His Arg Arg Trp Ser Glu Ala Gln Ile Ala Asp Asn Trp Ser Pro	
290 295 300	
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Phe 305	Gln	Arg	Met	Ala	Val 310	Glu	Phe	Phe	Glu	Asp 315	Cys	Arg	Val	Glu	Thr 320	
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Leu	Leu	Met	Arg	Glu	Tyr	Pro	Gly	Leu	Ala	Arg	Ile	Phe	Arg	Ala	Leu	
				325					330					335		
cac	cca	aag	cca	gtt	gaa	gct	gct	tgt	gac	ggg	gaa	act	act	tct	tgt	1056
His	Pro	Lys	Pro	Val	Glu	Ala	Ala	Cys	Asp	Gly	Glu	Thr	Thr	Ser	Cys	
			340					345					350			
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Leu	Arg	His	Arg	Leu	Ala	Met	Leu	Ser	Arg	Ala	Phe	Ile	Asp	Pro	Asp	
		355					360					365				
cac	ggg	tac	gct	gct	cca	gtt	ttg	aac	gac	ttc	gtt	gct	aga	ttc	cac	1152
His	Gly	Tyr	Ala	Ala	Pro	Val	Leu	Asn	Asp	Phe	Val	Ala	Arg	Phe	His	
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gct	aga	ttg	gct	gac	ggg	act	tct	tct	act	tct	gaa	atg	gct	gac	ttg	1200
Ala	Arg	Leu	Ala	Asp	Gly	Thr	Ser	Ser	Thr	Ser	Glu	Met	Ala	Asp	Leu	
385					390					395					400	
gct	ttg	tct	tac	gtt	gct	aag	act	aga	aga	cca	tct	gac	caa	ttc	gct	1248
Ala	Leu	Ser	Tyr	Val	Ala	Lys	Thr	Arg	Arg	Pro	Ser	Asp	Gln	Phe	Ala	
			405						410					415		
aag	gtt	cac	ttc	gac	gac	act	gtt	gtt	gac	tac	aga	gac	gac	aac	aga	1296
Lys	Val	His	Phe	Asp	Asp	Thr	Val	Val	Asp	Tyr	Arg	Asp	Asp	Asn	Arg	
		420					425						430			
caa	ttg	tg	aag	ttc	atc	gaa	gaa	ggg	gac	gaa	gaa	gaa	gct	ttc	gac	1344
Gln	Leu	Trp	Lys	Phe	Ile	Glu	Glu	Gly	Asp	Glu	Glu	Glu	Ala	Phe	Asp	
	435					440							445			
gct	aag	aga	aag	atc	gaa	cca	ggg	gaa	gaa	atc	caa	ggg	ttg	cca	cca	1392
Ala	Lys	Arg	Lys	Ile	Glu	Pro	Gly	Glu	Glu	Ile	Gln	Gly	Leu	Pro	Pro	
	450				455					460						
aga	cac	tac	cca	gaa	tg	gac	tac	act	tct	caa	act	tac	aga	cca	gac	1440
Arg	His	Tyr	Pro	Glu	Trp	Asp	Tyr	Thr	Ser	Gln	Thr	Tyr	Arg	Pro	Asp	
465				470						475				480		
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Trp	Val	Ser	Val	Tyr	Glu	Gly	Leu	His	Arg	Ser	Gly	Asn	Ala	Gly	Asp	
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Ile	Asp	Arg	Leu	Leu	Ala	Lys	His	Ala	Ala	Leu	Ala	Lys	Arg	Leu	Lys	
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Lys	Met	Leu	Asp	Leu	Leu	Lys	Pro	Gln	Asp	Lys	Val	Arg	Val	Arg	Tyr	
	515					520					525					
caa	gaa	gaa	ggg	tct	gaa	ttg	gac	ttg	gac	gtt	gct	atc	aga	tct	ttg	1632
Gln	Glu	Glu	Gly	Ser	Glu	Leu	Asp	Leu	Asp	Val	Ala	Ile	Arg	Ser	Leu	
	530				535					540						
atc	gac	ttc	aag	ggg	ggg	gct	act	cca	gac	cca	aga	atc	aac	atg	tct	1680
Ile	Asp	Phe	Lys	Gly	Gly	Ala	Thr	Pro	Asp	Pro	Arg	Ile	Asn	Met	Ser	
545				550						555				560		
cac	aga	tct	gac	ggg	aga	gac	atc	gct	gtt	atg	ttg	ttg	ttg	gac	ttg	1728
His	Arg	Ser	Asp	Gly	Arg	Asp	Ile	Ala	Val	Met	Leu	Leu	Leu	Asp	Leu	
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tct	gaa	tct	ttg	aac	gaa	aag	gct	gct	ggg	gct	ggg	caa	act	atc	ttg	1776
Ser	Glu	Ser	Leu	Asn	Glu	Lys	Ala	Ala	Gly	Ala	Gly	Gln	Thr	Ile	Leu	
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gaa	ttg	tct	caa	gaa	gct	gtt	tct	ttg	ttg	gct	tg	tct	atc	gaa	aag	1824
Glu	Leu	Ser	Gln	Glu	Ala	Val	Ser	Leu	Leu	Ala	Trp	Ser	Ile	Glu	Lys	
	595					600					605					
ttg	ggg	gac	cca	ttc	gct	atc	gct	ggg	ttc	cac	tct	aac	act	aga	cac	1872
Leu	Gly	Asp	Pro	Phe	Ala	Ile	Ala	Gly	Phe	His	Ser	Asn	Thr	Arg	His	
	610					615					620					

Pro	Glu	Arg	Gln	Gln	Asp	Tyr	Phe	Ser	Leu	Gln	Ser	Ala	Asp	Ala	Arg
			180				185						190		
Ala	Val	Leu	Gln	Arg	Glu	Arg	His	Gly	Thr	Leu	Leu	Val	Asp	Val	Glu
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Arg	Lys	Leu	Asp	Leu	Tyr	Leu	Arg	Gly	Leu	Trp	Gln	Asp	His	Asp	His
			210				215						220		
Leu	Val	Pro	Tyr	Ser	Thr	Ala	Phe	Asp	Glu	Ile	Arg	Lys	Pro	Val	Pro
			225				230						235		
Tyr	Tyr	Asp	Lys	Leu	Gly	Met	Arg	Leu	Pro	Asp	Val	Tyr	Asp	Asp	Leu
			245				250						255		
Val	Leu	Pro	Cys	Pro	Ala	Gly	Arg	Gly	Gly	Ala	Gly	Gly	Glu	Asp	Val
			260				265						270		
Leu	Val	Ser	Gly	Leu	Asp	Arg	Tyr	Arg	Ala	Thr	Leu	Ala	His	Met	Val
			275				280						285		
Gly	His	Arg	Arg	Trp	Ser	Glu	Ala	Gln	Ile	Ala	Asp	Asn	Trp	Ser	Pro
			290				295						300		
Phe	Gln	Arg	Met	Ala	Val	Glu	Phe	Phe	Glu	Asp	Cys	Arg	Val	Glu	Thr
			305				310						315		
Leu	Leu	Met	Arg	Glu	Tyr	Pro	Gly	Leu	Ala	Arg	Ile	Phe	Arg	Ala	Leu
			325				330						335		
His	Pro	Lys	Pro	Val	Glu	Ala	Ala	Cys	Asp	Gly	Glu	Thr	Thr	Ser	Cys
			340				345						350		
Leu	Arg	His	Arg	Leu	Ala	Met	Leu	Ser	Arg	Ala	Phe	Ile	Asp	Pro	Asp
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His	Gly	Tyr	Ala	Ala	Pro	Val	Leu	Asn	Asp	Phe	Val	Ala	Arg	Phe	His
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Ala	Arg	Leu	Ala	Asp	Gly	Thr	Ser	Ser	Thr	Ser	Glu	Met	Ala	Asp	Leu
			385				390						395		
Ala	Leu	Ser	Tyr	Val	Ala	Lys	Thr	Arg	Arg	Pro	Ser	Asp	Gln	Phe	Ala
			405				410						415		
Lys	Val	His	Phe	Asp	Asp	Thr	Val	Val	Asp	Tyr	Arg	Asp	Asp	Asn	Arg
			420				425						430		
Gln	Leu	Trp	Lys	Phe	Ile	Glu	Glu	Gly	Asp	Glu	Glu	Glu	Ala	Phe	Asp
			435				440						445		
Ala	Lys	Arg	Lys	Ile	Glu	Pro	Gly	Glu	Glu	Ile	Gln	Gly	Leu	Pro	Pro
			450				455						460		
Arg	His	Tyr	Pro	Glu	Trp	Asp	Tyr	Thr	Ser	Gln	Thr	Tyr	Arg	Pro	Asp
			465				470						475		
Trp	Val	Ser	Val	Tyr	Glu	Gly	Leu	His	Arg	Ser	Gly	Asn	Ala	Gly	Asp
			485				490						495		
Ile	Asp	Arg	Leu	Leu	Ala	Lys	His	Ala	Ala	Leu	Ala	Lys	Arg	Leu	Lys
			500				505						510		
Lys	Met	Leu	Asp	Leu	Leu	Lys	Pro	Gln	Asp	Lys	Val	Arg	Val	Arg	Tyr
			515				520						525		
Gln	Glu	Glu	Gly	Ser	Glu	Leu	Asp	Leu	Asp	Val	Ala	Ile	Arg	Ser	Leu
			530				535						540		
Ile	Asp	Phe	Lys	Gly	Gly	Ala	Thr	Pro	Asp	Pro	Arg	Ile	Asn	Met	Ser
			545				550						555		
His	Arg	Ser	Asp	Gly	Arg	Asp	Ile	Ala	Val	Met	Leu	Leu	Leu	Asp	Leu
			565				570						575		
Ser	Glu	Ser	Leu	Asn	Glu	Lys	Ala	Ala	Gly	Ala	Gly	Gln			

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595	600	605	
Leu Gly Asp Pro Phe Ala Ile Ala Gly Phe His Ser Asn Thr Arg His			
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Asp Val Arg Tyr Phe His Ile Lys Gly Tyr Ser Glu Arg Trp Asn Asp			
625	630	635	640
Asp Val Lys Ala Arg Leu Ala Ala Met Glu Ala Gly Tyr Ser Thr Arg			
645	650	655	
Met Gly Ala Ala Met Arg His Ala Ala His Tyr Leu Ser Ala Arg Pro			
660	665	670	
Ala Asp Lys Lys Leu Met Leu Ile Leu Thr Asp Gly Arg Pro Ser Asp			
675	680	685	
Val Asp Ala Ala Asp Glu Arg Leu Leu Val Glu Asp Ala Arg Gln Ala			
690	695	700	
Val Lys Glu Leu Asp Arg Gln Gly Ile Phe Ala Tyr Cys Ile Ser Leu			
705	710	715	720
Asp Ala Gln Leu Lys Ala Gly Ala Asp Asp Tyr Val Ala Glu Ile Phe			
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Gly Arg Gln Tyr Thr Val Ile Asp Arg Val Glu Arg Leu Pro Glu Arg			
740	745	750	
Leu Pro Glu Leu Phe Met Ala Leu Thr Lys			
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<210> SEQ ID NO 9			
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<220> FEATURE:			
<223> OTHER INFORMATION: synthetic, based on GroEL from E. coli - codon			
optimized			
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ttg aga ggt gtt aac gtt ttg gct gac gct gtt aag gtt act ttg ggt			96
Leu Arg Gly Val Asn Val Leu Ala Asp Ala Val Lys Val Thr Leu Gly			
20 25 30			
cca aag ggt aga aac gtt gtt ttg gac aag tct ttc ggt gct cca act			144
Pro Lys Gly Arg Asn Val Val Leu Asp Lys Ser Phe Gly Ala Pro Thr			
35 40 45			
atc act aag gac ggt gtt tct gtt gct aga gaa atc gaa ttg gaa gac			192
Ile Thr Lys Asp Gly Val Ser Val Ala Arg Glu Ile Glu Leu Glu Asp			
50 55 60			
aag ttc gaa aac atg ggt gct caa atg gtt aag gaa gtt gct tct aag			240
Lys Phe Glu Asn Met Gly Ala Gln Met Val Lys Glu Val Ala Ser Lys			
65 70 75 80			
gct aac gac gct gct ggt gac ggt act act act gct act gtt ttg gct			288
Ala Asn Asp Ala Ala Gly Asp Gly Thr Thr Thr Ala Thr Val Leu Ala			
85 90 95			
caa gct atc atc act gaa ggt ttg aag gct gtt gct gct ggt atg aac			336
Gln Ala Ile Ile Thr Glu Gly Leu Lys Ala Val Ala Ala Gly Met Asn			
100 105 110			
cca atg gac ttg aag aga ggt atc gac aag gct gtt act gct gct gtt			384
Pro Met Asp Leu Lys Arg Gly Ile Asp Lys Ala Val Thr Ala Ala Val			
115 120 125			
gaa gaa ttg aag gct ttg tct gtt cca tgt tct gac tct aag gct atc			432

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Glu 130	Glu	Leu	Lys	Ala	Leu	Ser 135	Val	Pro	Cys	Ser	Asp 140	Ser	Lys	Ala	Ile	
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Ala	Gln	Val	Gly	Thr	Ile	Ser	Ala	Asn	Ser	Asp	Glu	Thr	Val	Gly	Lys	
145					150					155					160	
ttg	atc	gct	gaa	gct	atg	gac	aag	ggt	ggt	aag	gaa	ggt	ggt	atc	act	528
Leu	Ile	Ala	Glu	Ala	Met	Asp	Lys	Val	Gly	Lys	Glu	Gly	Val	Ile	Thr	
				165					170					175		
ggt	gaa	gac	ggt	act	ggt	ttg	caa	gac	gaa	ttg	gac	ggt	ggt	gaa	ggt	576
Val	Glu	Asp	Gly	Thr	Gly	Leu	Gln	Asp	Glu	Leu	Asp	Val	Val	Glu	Gly	
			180					185					190			
atg	caa	ttc	gac	aga	ggt	tac	ttg	tct	cca	tac	ttc	atc	aac	aag	cca	624
Met	Gln	Phe	Asp	Arg	Gly	Tyr	Leu	Ser	Pro	Tyr	Phe	Ile	Asn	Lys	Pro	
		195					200					205				
gaa	act	ggt	gct	ggt	gaa	ttg	gaa	tct	cca	ttc	atc	ttg	ttg	gct	gac	672
Glu	Thr	Gly	Ala	Val	Glu	Leu	Glu	Ser	Pro	Phe	Ile	Leu	Leu	Ala	Asp	
	210					215					220					
aag	aag	atc	tct	aac	atc	aga	gaa	atg	ttg	cca	ggt	ttg	gaa	gct	ggt	720
Lys	Lys	Ile	Ser	Asn	Ile	Arg	Glu	Met	Leu	Pro	Val	Leu	Glu	Ala	Val	
225					230					235					240	
gct	aag	gct	ggt	aag	cca	ttg	ttg	atc	atc	gct	gaa	gac	ggt	gaa	ggt	768
Ala	Lys	Ala	Gly	Lys	Pro	Leu	Leu	Ile	Ile	Ala	Glu	Asp	Val	Glu	Gly	
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gaa	gct	ttg	gct	act	ttg	ggt	ggt	aac	act	atg	aga	ggt	atc	ggt	aag	816
Glu	Ala	Leu	Ala	Thr	Leu	Val	Val	Asn	Thr	Met	Arg	Gly	Ile	Val	Lys	
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ggt	gct	gct	ggt	aag	gct	cca	ggt	ttc	ggt	gac	aga	aga	aag	gct	atg	864
Val	Ala	Ala	Val	Lys	Ala	Pro	Gly	Phe	Gly	Asp	Arg	Arg	Lys	Ala	Met	
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ttg	caa	gac	atc	gct	act	ttg	act	ggt	ggt	act	ggt	atc	tct	gaa	gaa	912
Leu	Gln	Asp	Ile	Ala	Thr	Leu	Thr	Gly	Gly	Thr	Val	Ile	Ser	Glu	Glu	
		290					295				300					
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Ile	Gly	Met	Glu	Leu	Glu	Lys	Ala	Thr	Leu	Glu	Asp	Leu	Gly	Gln	Ala	
305					310					315					320	
aag	aga	ggt	ggt	atc	aac	aag	gac	act	act	act	atc	atc	gac	ggt	ggt	1008
Lys	Arg	Val	Val	Ile	Asn	Lys	Asp	Thr	Thr	Thr	Ile	Ile	Asp	Gly	Val	
			325						330					335		
ggt	gaa	gaa	gct	gct	atc	caa	ggt	aga	ggt	gct	caa	atc	aga	caa	caa	1056
Gly	Glu	Glu	Ala	Ala	Ile	Gln	Gly	Arg	Val	Ala	Gln	Ile	Arg	Gln	Gln	
			340					345					350			
atc	gaa	gaa	gct	act	tct	gac	tac	gac	aga	gaa	aag	ttg	caa	gaa	aga	1104
Ile	Glu	Glu	Ala	Thr	Ser	Asp	Tyr	Asp	Arg	Glu	Lys	Leu	Gln	Glu	Arg	
		355				360						365				
ggt	gct	aag	ttg	gct	ggt	ggt	gct	ggt	atc	aag	ggt	ggt	gct	gct		1152
Val	Ala	Lys	Leu	Ala	Gly	Gly	Val	Ala	Val	Ile	Lys	Val	Gly	Ala	Ala	
		370				375					380					
act	gaa	ggt	gaa	atg	aag	gaa	aag	aag	gct	aga	ggt	gaa	gac	gct	ttg	1200
Thr	Glu	Val	Glu	Met	Lys	Glu	Lys	Lys	Ala	Arg	Val	Glu	Asp	Ala	Leu	
		385			390					395					400	
cac	gct	act	aga	gct	gct	ggt	gaa	gaa	ggt	ggt	ggt	gct	ggt	ggt	ggt	1248
His	Ala	Thr	Arg	Ala	Ala	Val	Glu	Glu	Gly	Val	Val	Ala	Gly	Gly	Gly	
			405						410					415		
ggt	gct	ttg	atc	aga	ggt	gct	tct	aag	ttg	gct	gac	ttg	aga	ggt	caa	1296
Val	Ala	Leu	Ile	Arg	Val	Ala	Ser	Lys	Leu	Ala	Asp	Leu	Arg	Gly	Gln	
		420						425					430			
aac	gaa	gac	caa	aac	ggt	ggt	atc	aag	ggt	gct	ttg	aga	gct	atg	gaa	1344
Asn	Glu	Asp	Gln	Asn	Val	Gly	Ile	Lys	Val	Ala	Leu	Arg	Ala	Met	Glu	
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gct cca ttg aga caa atc gtt ttg aac tgt ggt gaa gaa cca tct gtt	1392
Ala Pro Leu Arg Gln Ile Val Leu Asn Cys Gly Glu Glu Pro Ser Val	
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ggt gct aac act gtt aag ggt ggt gac ggt aac tac ggt tac aac gct	1440
Val Ala Asn Thr Val Lys Gly Gly Asp Gly Asn Tyr Gly Tyr Asn Ala	
465 470 475 480	
gct act gaa gaa tac ggt aac atg atc gac atg ggt atc ttg gac cca	1488
Ala Thr Glu Glu Tyr Gly Asn Met Ile Asp Met Gly Ile Leu Asp Pro	
485 490 495	
act aag gtt act aga tct gct ttg caa tac gct gct tct gtt gct ggt	1536
Thr Lys Val Thr Arg Ser Ala Leu Gln Tyr Ala Ala Ser Val Ala Gly	
500 505 510	
ttg atg atc act act gaa tgt atg gtt act gac ttg cca aag aac gac	1584
Leu Met Ile Thr Thr Glu Cys Met Val Thr Asp Leu Pro Lys Asn Asp	
515 520 525	
gct gct gac ttg ggt gct gct ggt ggt atg ggt ggt atg ggt ggt atg	1632
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<210> SEQ ID NO 10

<211> LENGTH: 548

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 10

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Pro Lys Gly Arg Asn Val Val Leu Asp Lys Ser Phe Gly Ala Pro Thr	
35 40 45	
Ile Thr Lys Asp Gly Val Ser Val Ala Arg Glu Ile Glu Leu Glu Asp	
50 55 60	
Lys Phe Glu Asn Met Gly Ala Gln Met Val Lys Glu Val Ala Ser Lys	
65 70 75 80	
Ala Asn Asp Ala Ala Gly Asp Gly Thr Thr Thr Ala Thr Val Leu Ala	
85 90 95	
Gln Ala Ile Ile Thr Glu Gly Leu Lys Ala Val Ala Ala Gly Met Asn	
100 105 110	
Pro Met Asp Leu Lys Arg Gly Ile Asp Lys Ala Val Thr Ala Ala Val	
115 120 125	
Glu Glu Leu Lys Ala Leu Ser Val Pro Cys Ser Asp Ser Lys Ala Ile	
130 135 140	
Ala Gln Val Gly Thr Ile Ser Ala Asn Ser Asp Glu Thr Val Gly Lys	
145 150 155 160	
Leu Ile Ala Glu Ala Met Asp Lys Val Gly Lys Glu Gly Val Ile Thr	
165 170 175	
Val Glu Asp Gly Thr Gly Leu Gln Asp Glu Leu Asp Val Val Glu Gly	
180 185 190	
Met Gln Phe Asp Arg Gly Tyr Leu Ser Pro Tyr Phe Ile Asn Lys Pro	
195 200 205	
Glu Thr Gly Ala Val Glu Leu Glu Ser Pro Phe Ile Leu Leu Ala Asp	
210 215 220	

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Lys Lys Ile Ser Asn Ile Arg Glu Met Leu Pro Val Leu Glu Ala Val
 225 230 235 240
 Ala Lys Ala Gly Lys Pro Leu Leu Ile Ile Ala Glu Asp Val Glu Gly
 245 250 255
 Glu Ala Leu Ala Thr Leu Val Val Asn Thr Met Arg Gly Ile Val Lys
 260 265 270
 Val Ala Ala Val Lys Ala Pro Gly Phe Gly Asp Arg Arg Lys Ala Met
 275 280 285
 Leu Gln Asp Ile Ala Thr Leu Thr Gly Gly Thr Val Ile Ser Glu Glu
 290 295 300
 Ile Gly Met Glu Leu Glu Lys Ala Thr Leu Glu Asp Leu Gly Gln Ala
 305 310 315 320
 Lys Arg Val Val Ile Asn Lys Asp Thr Thr Thr Ile Ile Asp Gly Val
 325 330 335
 Gly Glu Glu Ala Ala Ile Gln Gly Arg Val Ala Gln Ile Arg Gln Gln
 340 345 350
 Ile Glu Glu Ala Thr Ser Asp Tyr Asp Arg Glu Lys Leu Gln Glu Arg
 355 360 365
 Val Ala Lys Leu Ala Gly Gly Val Ala Val Ile Lys Val Gly Ala Ala
 370 375 380
 Thr Glu Val Glu Met Lys Glu Lys Lys Ala Arg Val Glu Asp Ala Leu
 385 390 395 400
 His Ala Thr Arg Ala Ala Val Glu Glu Gly Val Val Ala Gly Gly Gly
 405 410 415
 Val Ala Leu Ile Arg Val Ala Ser Lys Leu Ala Asp Leu Arg Gly Gln
 420 425 430
 Asn Glu Asp Gln Asn Val Gly Ile Lys Val Ala Leu Arg Ala Met Glu
 435 440 445
 Ala Pro Leu Arg Gln Ile Val Leu Asn Cys Gly Glu Glu Pro Ser Val
 450 455 460
 Val Ala Asn Thr Val Lys Gly Gly Asp Gly Asn Tyr Gly Tyr Asn Ala
 465 470 475 480
 Ala Thr Glu Glu Tyr Gly Asn Met Ile Asp Met Gly Ile Leu Asp Pro
 485 490 495
 Thr Lys Val Thr Arg Ser Ala Leu Gln Tyr Ala Ala Ser Val Ala Gly
 500 505 510
 Leu Met Ile Thr Thr Glu Cys Met Val Thr Asp Leu Pro Lys Asn Asp
 515 520 525
 Ala Ala Asp Leu Gly Ala Ala Gly Gly Met Gly Gly Met Gly Gly Met
 530 535 540
 Gly Gly Met Met
 545

<210> SEQ ID NO 11
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 <213> ORGANISM: Artificial Sequence
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 <220> FEATURE:
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Met	Asn	Ile	Arg	Pro	Leu	His	Asp	Arg	Val	Ile	Val	Lys	Arg	Lys	Glu	
1				5					10					15		
ggt	gaa	act	aag	tct	gct	ggg	ggg	atc	ggt	ttg	act	ggg	tct	gct	gct	96
Val	Glu	Thr	Lys	Ser	Ala	Gly	Gly	Ile	Val	Leu	Thr	Gly	Ser	Ala	Ala	
			20					25				30				
gct	aag	tct	act	aga	ggg	gaa	ggt	ttg	gct	ggt	ggg	aac	ggg	aga	atc	144
Ala	Lys	Ser	Thr	Arg	Gly	Glu	Val	Leu	Ala	Val	Gly	Asn	Gly	Arg	Ile	
			35				40				45					
ttg	gaa	aac	ggg	gaa	ggt	aag	cca	ttg	gac	ggt	aag	ggt	ggg	gac	atc	192
Leu	Glu	Asn	Gly	Glu	Val	Lys	Pro	Leu	Asp	Val	Lys	Val	Gly	Asp	Ile	
	50					55				60						
ggt	atc	ttc	aac	gac	ggg	tac	ggg	ggt	aag	tct	gaa	aag	atc	gac	aac	240
Val	Ile	Phe	Asn	Asp	Gly	Tyr	Gly	Val	Lys	Ser	Glu	Lys	Ile	Asp	Asn	
65					70				75					80		
gaa	gaa	ggt	ttg	atc	atg	tct	gaa	tct	gac	atc	ttg	gct	atc	ggt	gaa	288
Glu	Glu	Val	Leu	Ile	Met	Ser	Glu	Ser	Asp	Ile	Leu	Ala	Ile	Val	Glu	
			85						90					95		
gct	taa															294
Ala																

<210> SEQ ID NO 12
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 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 12

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Val	Glu	Thr	Lys	Ser	Ala	Gly	Gly	Ile	Val	Leu	Thr	Gly	Ser	Ala	Ala	
			20					25				30				
Ala	Lys	Ser	Thr	Arg	Gly	Glu	Val	Leu	Ala	Val	Gly	Asn	Gly	Arg	Ile	
			35				40				45					
Leu	Glu	Asn	Gly	Glu	Val	Lys	Pro	Leu	Asp	Val	Lys	Val	Gly	Asp	Ile	
	50					55				60						
Val	Ile	Phe	Asn	Asp	Gly	Tyr	Gly	Val	Lys	Ser	Glu	Lys	Ile	Asp	Asn	
65					70				75					80		
Glu	Glu	Val	Leu	Ile	Met	Ser	Glu	Ser	Asp	Ile	Leu	Ala	Ile	Val	Glu	
			85						90					95		
Ala																

<210> SEQ ID NO 13
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
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<400> SEQUENCE: 13

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<210> SEQ ID NO 14
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 <220> FEATURE:
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<210> SEQ ID NO 15
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 agctggagct cagtttatca ttatc 85

<210> SEQ ID NO 16
 <211> LENGTH: 81
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: primer

 <400> SEQUENCE: 16

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 ggtaccggcc gcaaattaaa g 81

<210> SEQ ID NO 17
 <211> LENGTH: 120
 <212> TYPE: DNA
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 <220> FEATURE:
 <223> OTHER INFORMATION: primer

 <400> SEQUENCE: 17

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 ccagcttttg ttcccttttag tgagggttaa ttgcgcgctt ggcgtaatca tggtcatagc 120

<210> SEQ ID NO 18
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
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 gacatatcag catacatggc tatggcacg cagacattcc aggcatacta cacagattcc 60
 atcactctta ccaggctagg acgaccctac tcatgtattg agatcgacga gatttctagg 120

<210> SEQ ID NO 19
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 <212> TYPE: DNA
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 cttaactcct gtaaaaacaa aaaaaaaaaa aggcatagca agctggagct cagtttatc 119

<210> SEQ ID NO 20
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
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<400> SEQUENCE: 20

agatatactg caaagtcg agcaacagtc gtataactcg agcagccctc tactttgttg 60

ttgcgctaag agaattggacc 80

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

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ttgcgctaag agaattggacc 80

<210> SEQ ID NO 22

<211> LENGTH: 81

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

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gctggagctc tagtacggat t 81

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<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: primer

<400> SEQUENCE: 23

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gtaccggccg caaattaaag 80

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

<400> SEQUENCE: 24

gacatatcag catacatggc tatgg 25

<210> SEQ ID NO 25

<211> LENGTH: 26

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

<400> SEQUENCE: 25

ggacacgctt gacagaatgt caaagg 26

<210> SEQ ID NO 26

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer

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 <212> TYPE: DNA
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 <220> FEATURE:
 <223> OTHER INFORMATION: primer

<400> SEQUENCE: 27

cctagaaatc tcgtcgatct c 21

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 <211> LENGTH: 20
 <212> TYPE: DNA
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 <223> OTHER INFORMATION: primer

<400> SEQUENCE: 28

atcactctta ccaggctagg 20

<210> SEQ ID NO 29
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 <212> TYPE: DNA
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 <220> FEATURE:
 <223> OTHER INFORMATION: primer

<400> SEQUENCE: 29

ctggacctta atcgtgtgcg catcctc 27

<210> SEQ ID NO 30
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 <212> TYPE: DNA
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 <220> FEATURE:
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<400> SEQUENCE: 30

ccgtatagct taatagccag ctttacc 27

<210> SEQ ID NO 31
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: primer

<400> SEQUENCE: 31

gctatgacca tgattacgcc aagc 24

<210> SEQ ID NO 32
 <211> LENGTH: 80
 <212> TYPE: DNA
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 <220> FEATURE:
 <223> OTHER INFORMATION: primer

<400> SEQUENCE: 32

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ctgtgaagat cccagcaaag 80

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 33

agctcattga tcccttaaac ttctttttcg gtgtatgact tatgaggggtg agaatgcgaa 60
atggcgtgga aatgtgatca aaggtaataa aacgtcatat atccgcaggc taaccggaac 120

<210> SEQ ID NO 34
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 34

ggcgattaag ttgggtaacg 20

<210> SEQ ID NO 35
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: primer

<400> SEQUENCE: 35

aactgagctc cagctgtacc 20

<210> SEQ ID NO 36
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 36

acgcgtgtac gcatgtaac 19

<210> SEQ ID NO 37
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 37

gcgcgtggct tcctataatc 20

<210> SEQ ID NO 38
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 38

gtgaatgctg gtcgctatac 20

<210> SEQ ID NO 39
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 39

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<210> SEQ ID NO 40
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 40

acctgaccta caggaaagag 20

<210> SEQ ID NO 41
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 41

tgaagtggta cggcgatgc 19

<210> SEQ ID NO 42
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 42

atagccaccc aaggcatttc 20

<210> SEQ ID NO 43
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 43

ccgcactttc tccatgagg 19

<210> SEQ ID NO 44
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 44

cgacggttac ggtgtaag 19

<210> SEQ ID NO 45
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer

<400> SEQUENCE: 45

cttcggctc ctatgtgtg 20

What is claimed is:

1. A recombinant yeast cell comprising:
one or more recombinant heterologous, nucleic acid sequences encoding Ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco) and phosphoribulokinase (PRK), wherein the amino acid sequence of the Rubisco has at least 80% sequence identity with the sequence of SEQ ID NO: 2, and
the amino acid sequence of the PRK has at least 70% sequence identity with the sequence of SEQ ID NO: 4; and
nucleic acid sequences encoding a GroEL or a functional homologue of GroEL, and a GroES or a functional homologue of GroES, wherein the respective GroEL and GroES homologues are functional as molecular chaperones,
wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 70% sequence identity with the sequence of SEQ ID NO: 10; and
wherein the amino acid sequence of the GroES or the functional homologue of GroES has at least 70% sequence identity with the sequence of SEQ ID NO: 12.
2. The recombinant yeast cell of claim 1, wherein the Rubisco uses CO₂ as an electron acceptor.
3. The recombinant yeast cell of claim 1, wherein the Rubisco is a single subunit Rubisco.
4. The recombinant yeast cell of claim 1, wherein the Rubisco is a prokaryotic form-II Rubisco.
5. The recombinant yeast cell of claim 1, wherein the PRK is a PRK originating from a eukaryote.
6. The recombinant yeast cell of claim 5, wherein the PRK originates from a Caryophyllales plant.
7. The recombinant yeast cell of claim 6, wherein the Caryophyllales plant is *Amaranthaceae* or *Spinacia*.
8. The recombinant yeast cell of claim 1, wherein the genus of said yeast cell is selected from the group consisting of *Saccharomycesaceae*, *Schizosaccharomyces*, *Torulaspora*, *Kluyveromyces*, *Pichia*, *Zygosaccharomyces*, *Brettanomyces*, *Metschnikowia*, *Issatchenkia*, *Kloeckera*, and *Aureobasidium*.
9. The recombinant yeast cell of claim 8, wherein the genus of yeast cell is *Saccharomycesaceae*.
10. The recombinant yeast cell of claim 9, wherein the yeast cell is selected from the group consisting of *Saccharomyces cerevisiae*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum*, and *Saccharomyces bayanus*.
11. A method for preparing an alcohol under anaerobic conditions comprising fermenting a carbon source and the recombinant yeast cell of claim 1.
12. The recombinant yeast cell of claim 7, wherein the PRK is a *Spinacia* PRK.
13. The recombinant yeast cell of claim 9, wherein the yeast cell is a *Saccharomyces cerevisiae*.
14. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 75% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence

of the GroES or the functional homologue of GroES has at least 75% sequence identity with the sequence of SEQ ID NO: 12.

15. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 80% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 80% sequence identity with the sequence of SEQ ID NO: 12.

16. The recombinant yeast cell of claim 1, further comprising a heterologous nucleic acid sequence encoding a dehydrogenase.

17. The recombinant yeast cell of claim 16, wherein the dehydrogenase is a xylitol dehydrogenase.

18. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 85% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 85% sequence identity with the sequence of SEQ ID NO: 12.

19. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 90% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 90% sequence identity with the sequence of SEQ ID NO: 12.

20. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 95% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 95% sequence identity with the sequence of SEQ ID NO: 12.

21. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 85% sequence identity with the sequence of SEQ ID NO: 2.

22. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 90% sequence identity with the sequence of SEQ ID NO: 2.

23. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 95% sequence identity with the sequence of SEQ ID NO: 2.

24. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 75% sequence identity with the sequence of SEQ ID NO: 4.

25. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 80% sequence identity with the sequence of SEQ ID NO: 4.

26. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 85% sequence identity with the sequence of SEQ ID NO: 4.

27. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 90% sequence identity with the sequence of SEQ ID NO: 4.

28. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 95% sequence identity with the sequence of SEQ ID NO: 4.

* * * * *